On JEPA Isotropy

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

We study the empirical risk minimization (ERM) of linear Joint Embedding Predictive Architecture (JEPA) and establish its key conditions for optimality. By casting the embedding bottleneck as a rank constraint, we formulate JEPA as a reduced-rank regression problem. This induces a risk decomposition into an irreducible error and an excess-risk bound. Within the irreducible term, we find the rank of the optimal predictor grows with target heterogeneity and saturates at the bottleneck. Within the excess-risk bound, we find spectral conditioning of the whitened end-to-end map governs minimization. Together, these conditions prescribe an isotropic regularizer. Under this regularization, we show such JEPA isotropy subsumes canonical JEPA isotropic embedding designs and hence justifies their empirical successes via ERM. Numerical experiments corroborate our theory.

1 Introduction

We study the empirical risk minimization (ERM) of Joint-Embedding Predictive Architecture (JEPA) [1, 2] and establish its conditions for optimality. JEPA jointly trains a context encoder, a target encoder, and a predictor so that the predicted target embedding matches the target's. It has become a prominent self-supervised learning paradigm [2–5]. A key recipe in JEPA is to enforce isotropy on the encoder's marginal embedding, either through architectures [2–4] or through distributional regularizations [6, 5]. Prior work justifies this recipe by appealing to typical self-supervised methods [7–9], by showing it prevents representational [10, 11] and early-training [12] collapse, or by scoping to specific objectives [5, 13]. More fundamentally, they regularize only the encoder and ignore the predictor, the defining component of JEPA beyond plain Joint-Embedding Architectures (JEAs). As a result, it remains unclear what JEPA's loss is optimizing. This work targets these gaps through a finite-sample analysis. Specifically, we answer:

- (Q1) What condition on the encoder and predictor *jointly* minimizes JEPA's loss?
- (Q2) How does the rank of the optimal predictor depend on the number and structure of targets?
- (Q3) Why does encoder isotropy work, and what unifies its existing forms?

Casting the embedding bottleneck as a rank constraint, we formulate JEPA as reduced-rank regression [14, 15] and decompose its finite-sample risk into an irreducible population term and an excess term (Theorem 3.15). We answer (Q1) by showing that the excess term is governed by a single condition on the end-to-end mapping. The excess term is minimized at a flat spectrum condition termed *predictive isotropy* (Lemma 4.1 and Theorem 4.2). We answer (Q2) by characterizing the irreducible term through the rank of the optimal predictor. It grows with the heterogeneity of targets and saturates at the embedding bottleneck (Propositions 3.4, 3.8 and 3.9).

We thus answer (Q3) from a gauge-invariant risk-minimization view. The predictive isotropy condition for (Q1) acts on the end-to-end mapping. Its SVD supplies a factorization into encoder and predictor, and gauge transformations reparameterize them within the same composition. The risk is gauge-invariant, so any factorization that realizes predictive isotropy minimizes it. Whitening, VICReg, Gaussian-embedding, and LeJEPA [16, 6, 13, 5] each select a gauge toward a specific marginal embedding (Theorems 4.5 and 4.6). Predictive isotropy subsumes and justifies encoder isotropy.

Building on this unification, we derive a predictive-isotropy regularizer that acts only on the end-to-end mapping from a maximum-entropy principle. Empirically, it improves both test risk and train-test gap over encoder isotropy on MNIST image prediction. As the gauge-invariant view for (Q3) predicts, it yields an isotropic-Gaussian embedding as a byproduct. Synthetic experiments further validate the intermediate claims of our analysis.

Contribution. Our contributions include:

- **First ERM Theory of JEPA.** We provide, to our knowledge, the first finite-sample theory of the JEPA objective. Under linearization, JEPA reduces to reduced-rank regression, yielding a risk decomposition governed by the spectrum of the whitened end-to-end map T_K (Theorem 3.15).
- **Identifying the Risk Minimizer.** We show that the excess-risk bound is uniquely minimized when T_K has a flat nonzero singular spectrum. This identifies predictive isotropy as an optimality condition for JEPA beyond the scope of existing encoder-only designs (Theorem 4.2).
- **Justification and Unification of Encoder Isotropy.** Our framework gives a unified risk-based explanation for empirically successful encoder-isotropy recipes, including Whitening, VICReg, Gaussian-embedding, and LeJEPA. Under predictive isotropy, these methods arise as gauge choices in the encoder-predictor factorization of T_K (Theorems 4.5 and 4.6).
- **Regularization Toward the Risk Optimum.** We translate the unique minimizer into a practical regularizer on the end-to-end mapping (18). Empirically, it reduces test risk and train-test gap, while inducing isotropic-Gaussian embedding as a natural byproduct rather than a constraint.

Roadmap. Section 2 introduces preliminaries. Section 3 develops the reduced-rank regression formulation and risk decomposition. Section 4 establishes predictive isotropy as the unique optimality condition and shows it subsumes existing encoder-isotropy designs. Section 5 reports numerical validations. Section 6 concludes. Section B, Section C, and Section D contain extended discussion and related work, full proofs, and experimental details, respectively.

2 Preliminaries

In this section, we introduce the notation and background needed for our JEPA analysis.

Let $z \in \mathbb{R}^D$ denote a data sample drawn from a distribution $\mathcal{P}$ on $\mathbb{R}^D$. A *mask* is an index set $m \subseteq [D]$ selecting a subset of coordinates. A *mask distribution* $\mathcal{D}$ is a probability distribution over tuples $(m_c, m_{t,1}, \dots, m_{t,K})$ specifying one *context block* and $K \geq 1$ *target blocks*.[1] For a sampled tuple, write $z_{\text{ctx}} := z_{m_c} \in \mathbb{R}^{D_x}$ and $z_{\text{tgt}}^{(k)} := z_{m_{t,k}} \in \mathbb{R}^{D_y^{(k)}}$ for the corresponding context and target subvectors. JEPA jointly trains an encoder and a predictor to predict the embedding of each target block from the embedding of the context block.

Architecture. JEPA consists of three components:

(i) A *context encoder* $f_\theta : \mathbb{R}^{D_x} \to \mathbb{R}^d$ mapping the context block to an embedding $x = f_\theta(z_{\text{ctx}})$;

(ii) A *target encoder* (*teacher*) $f_{\bar{\theta}} : \mathbb{R}^{D_y^{(k)}} \to \mathbb{R}^d$ mapping each target to an embedding $y^{(k)} = f_{\bar{\theta}}(z_{\text{tgt}}^{(k)})$;

(iii) A *predictor* $g_W : \mathbb{R}^d \times \mathcal{Z} \to \mathbb{R}^d$ that predicts the k-th target embedding $y^{(k)}$ from the context embedding x and target-side metadata $q_k \in \mathcal{Z}$, i.e., $\widehat{y}^{(k)} = g_W(x, q_k)$.

Throughout this work, we assume a frozen teacher and a linear end-to-end map.

Assumption 2.1 (Frozen Teacher)**.** The target encoder $f_{\bar{\theta}}$ is fixed during training of the context encoder f_θ and predictor g_W.

Assumption 2.2 (End-to-end linearity)**.** The end-to-end map $g_W \circ f_\theta : \mathbb{R}^{D_x} \to \mathbb{R}^{Kd}$ is linear.

For explicit matrix-level analysis (Section 3.1-3.5), we consider a specialized linearity assumption.

Assumption 2.3 (Linear factorization)**.** The context encoder is linear, and for each target block $k \in [K]$ with fixed metadata q_k, the predictor is linear in the context embedding:

$$f_\theta(z_{\text{ctx}}) = A z_{\text{ctx}}, \quad g_W(x, q_k) = W_k x, \qquad A \in \mathbb{R}^{d \times D_x}, \quad W_k \in \mathbb{R}^{d \times d}.$$

Note Assumption 2.3 implies Assumption 2.2 and serves as the working factorization. Section 4 relaxes it to nonlinear factorizations within the same end-to-end linear map (Lemma 4.4).

[1] We call a mask a *block mask* when it selects a contiguous index set. The target blocks need not be disjoint.

Under Assumption 2.3, write $Z := z_{\mathrm{ctx}}$ and regard the learned predictor from the context block to the stacked target embedding as $B := WA \in \mathbb{R}^{Kd\times D_x}$. Since B factors through the d-dimensional embedding, $\mathrm{rank}(B) \le d$. Conversely, every rank-r matrix B with $r \le d$ admits a factorization $B = WA$ with $A \in \mathbb{R}^{d\times D_x}$ and $W \in \mathbb{R}^{Kd\times d}$. Thus, for the linear frozen-teacher analysis, optimizing over encoder-predictor pairs (W, A) is equivalent to optimizing over rank-constrained maps B. Under Assumption 2.3, stacking the target embeddings,

$$Y := \left[(y^{(1)})^\top \quad \cdots \quad (y^{(K)})^\top\right]^\top \in \mathbb{R}^{Kd}, \tag{1}$$

$$\widehat{Y} = \begin{bmatrix} \widehat{y}^{(1)} \\ \vdots \\ \widehat{y}^{(K)} \end{bmatrix} = Wx, \qquad W := \begin{bmatrix} W_1 \\ \vdots \\ W_K \end{bmatrix} \in \mathbb{R}^{Kd\times d}. \tag{2}$$

Thus the rank-d end-to-end map from context to predicted target embeddings is $WA \in \mathbb{R}^{Kd\times D_x}$.

Below we define the JEPA training objective. Unless otherwise stated, all expectations are taken over $z \sim \mathcal{P}$ and $(m_c, m_{t,1}, \ldots, m_{t,K}) \sim \mathcal{D}$.

Definition 2.4 (JEPA's loss)**.** Write $Z := z_{\mathrm{ctx}}$, stack the target embeddings $y^{(k)} := f_{\bar\theta}(z^{(k)}_{\mathrm{tgt}})$ into Y as in (1), and the predictor blocks W_k into W as in (2). The JEPA's loss is, for any map $B \in \mathbb{R}^{Kd\times D_x}$,

$$\mathcal{L}_K(B) := \frac{1}{K}\,\mathbb{E}\big[\|Y - BZ\|^2\big], \tag{3}$$

and we write $\mathcal{L}_K(W, A) = \mathcal{L}_K(WA)$ for the loss of an encoder–predictor pair.

The composition WA acts on the context Z only through its row space, a low-dimensional subspace of $\mathbb{R}^{D_x}$ that captures the directions JEPA effectively operates on. We name it next.

Definition 2.5 (Predictive subspace)**.** $\mathcal{V}$ is a *predictive subspace* if $\mathcal{V} = \mathrm{row}(WA)$ for some predictor WA under Assumption 2.3. We denote by $\mathrm{Gr}(r, D_x)$ the Grassmannian $\{\mathcal{V} \subseteq \mathbb{R}^{D_x} : \dim \mathcal{V} = r\}$.

Denote the second moments $\Sigma_{ZZ} := \mathbb{E}[ZZ^\top] \in \mathbb{R}^{D_x\times D_x}$, $\Sigma_{Y^{(k)}Z} := \mathbb{E}[y^{(k)}Z^\top] \in \mathbb{R}^{d\times D_x}$ for $k \in [K]$, and their stack $\Sigma_{YZ} := \mathbb{E}[YZ^\top] = [\Sigma^\top_{Y^{(1)}Z}, \ldots, \Sigma^\top_{Y^{(K)}Z}]^\top \in \mathbb{R}^{Kd\times D_x}$. Assume $\Sigma_{ZZ} \succ 0$ for the reduced-rank analysis.

3 Main Theory

The section analyses JEPA's risk in two stages: what it is, and what minimizes it. Sections 3.1 to 3.3 show how target structure and the embedding bottleneck saturate the population risk; Section 3.4 identifies a condition as the residual lever beyond saturation; Section 3.5 unifies both into a risk decomposition. Sections 4 and 4.2 identify its unique minimizer and translate it into a regularizer.

3.1 JEPA as Reduced-Rank Regression

We show that, under the frozen-teacher and linearity (Assumptions 2.1 and 2.3), JEPA reduces to a reduced-rank regression problem. For $r \le d$, consider the rank-constrained JEPA problem

$$B^\star_r \in \arg\min_{B\in\mathbb{R}^{Kd\times D_x}} \mathcal{L}_K(B) \qquad \text{subject to} \qquad \mathrm{rank}(B) \le r, \tag{4}$$

where $\mathcal{L}_K(B)$ is the JEPA loss in (3). Recall Σ_{ZZ}, Σ_{YZ} from Section 2 and define

$$\underbrace{B^{\mathrm{raw}}_{\mathrm{OLS}} := \Sigma_{YZ}\Sigma_{ZZ}^{-1}}_{\text{unconstrained OLS map}}, \qquad \underbrace{T_K := B^{\mathrm{raw}}_{\mathrm{OLS}}\Sigma_{ZZ}^{1/2} = \Sigma_{YZ}\Sigma_{ZZ}^{-1/2}}_{\text{whitened end-to-end map}}. \tag{5}$$

Proposition 3.1 (JEPA as reduced-rank regression)**.** *Under Assumptions 2.1 and 2.3, the optimization problem* (4) *is exactly the reduced-rank regression (Definition A.1) with $p = D_x$, $q = Kd$, covariate $Z \in \mathbb{R}^{D_x}$, and response $Y \in \mathbb{R}^{Kd}$. Consequently, by Lemma A.2 its closed-form solution is*

$$B^\star_r = B^{\mathrm{raw}}_{\mathrm{OLS}}\Sigma_{ZZ}^{1/2}V_{K,r}V_{K,r}^\top\Sigma_{ZZ}^{-1/2}, \tag{6}$$

where $V_{K,r}$ contains the top-r right singular vectors of T_K.

Below we study when increasing the number of target blocks can enlarge the predictive structure accessible from the context.

3.2 Multi-Target Masking and Rank Growth

The JEPA predictor in (4) is subject to two rank bottlenecks: the explicit constraint $r \le d$, and an intrinsic rank induced by the data distribution. Lemma 3.2 makes the latter precise.

Lemma 3.2 (Rank of the JEPA solution)**.** *Let $B_r^\star$ be the solution to Proposition 3.1. Then*

$$\operatorname{rank}(B_r^\star) = \min\{r, \operatorname{rank}(T_K)\}. \tag{7}$$

Proof. See Section C.1 for a detailed proof. □

Lemma 3.2 identifies $\operatorname{rank}(T_K)$ as the intrinsic predictive rank: adding target blocks enlarges the predictive structure only insofar as it raises this rank. To study how $\operatorname{rank}(T_K)$ depends on the choice and number of target blocks, define the whitened block operators $M_k := \Sigma_{Y^{(k)}Z}\Sigma_{ZZ}^{-1/2}$ for $k \in [K]$, and aggregate them through the predictive Gram matrix

$$H_K := T_K^\top T_K = \Sigma_{ZZ}^{-1/2}\Sigma_{ZY}\Sigma_{YZ}\Sigma_{ZZ}^{-1/2} = \sum_{k=1}^{K} M_k^\top M_k \in \mathbb{R}^{D_x \times D_x}. \tag{8}$$

Each summand $M_k^\top M_k$ is positive semidefinite, so adding a block can only enlarge the range of H_K, equivalently shrinking its kernel. Throughout, we write $\lambda_i(\cdot)$ and $\sigma_i(\cdot)$ for the i-th largest eigenvalue and singular value, and $\|\cdot\|_{\mathrm{op}}$ for the operator norm. By (8), $\lambda_i(H_K) = \sigma_i(T_K)^2$ for every i.

We introduce a predictive rank (Definition 3.3, Proposition 3.4, and Corollary 3.5) measuring *how large* the predictable subspace is, and a heterogeneity level (Definition 3.6 and Proposition 3.8) measuring *how well* it can be recovered from samples.

Definition 3.3 (Predictive rank)**.** For a given K, the *predictive rank* is $r_{\mathrm{pop}}(K) := \operatorname{rank}(T_K)$.

As $T_K \in \mathbb{R}^{Kd \times D_x}$, the predictive rank obeys $r_{\mathrm{pop}}(K) \le \min\{Kd, D_x\}$. Intuitively, $r_{\mathrm{pop}}(K)$ is the predictive subspace dimension before the architectural truncation. We study how it depends on K.

Proposition 3.4 (Rank growth)**.** *Let $H_K := T_K^\top T_K$ and $r_{\mathrm{pop}}(K) := \operatorname{rank}(T_K)$ for $K \ge 1$. Then:*

1. ***Monotonicity.*** $r_{\mathrm{pop}}(K+1) \ge r_{\mathrm{pop}}(K)$.
2. ***Strict Growth under Complementarity.*** $r_{\mathrm{pop}}(K+1) > r_{\mathrm{pop}}(K)$ *if* $\ker(H_K) \not\subseteq \ker(M_{K+1})$.

Proof. See Section C.2 for a detailed proof. □

Corollary 3.5 (Saturation)**.** *There exists $K^\star$ with $r_{\mathrm{pop}}(K) = r_{\mathrm{pop}}(K^\star) \le D_x$ for all $K \ge K^\star$.*

Besides the predictive rank, it is useful to track how far H_K is from being rank-deficient on its range.

Definition 3.6 (Heterogeneity level)**.** The heterogeneity level λ_{het} is the smallest nonzero eigenvalue of H_K, i.e., $\lambda_{\mathrm{het}} := \lambda_{r_{\mathrm{pop}}(K)}(H_K)$.

Since $H_K = T_K^\top T_K$, the square root of λ_{het} is the smallest nonzero singular value of T_K:

$$\sigma_{r_{\mathrm{pop}}(K)}(T_K) = \sqrt{\lambda_{\mathrm{het}}}. \tag{9}$$

Remark 3.7 (Heterogeneity as complementarity). The rank strictly increases at step $K+1$ if and only if the new whitened block operator M_{K+1} acts nontrivially on $\ker(H_K)$, i.e., contributes a predictive direction outside the span accumulated from the first K blocks. Heterogeneity is therefore complementarity among the blockwise operators rather than their number.

The next proposition shows that λ_{het} improves how well the predictive subspace of T_K is identified.

Proposition 3.8 (Heterogeneity controls predictive-subspace identifiability)**.** *Assume the family $\{M_k\}_{k=1}^K$ has heterogeneity level $\lambda_{\mathrm{het}} > 0$ (Definition 3.6). Let $\widehat{T}_K$ be an empirical whitened operator and let $\mathcal{V}_{K,r_{\mathrm{pop}}}, \widehat{\mathcal{V}}_{K,r_{\mathrm{pop}}}$ denote the top-$r_{\mathrm{pop}}(K)$ right singular subspaces of T_K and $\widehat{T}_K$, respectively. Suppose the event $E_T^{\mathrm{abs}} := \{\|\widehat{T}_K - T_K\|_{\mathrm{op}} \le \eta_n^T(\delta)\}$ satisfies $\mathbb{P}(E_T^{\mathrm{abs}}) \ge 1-\delta$ for some $\eta_n^T(\delta) < \sqrt{\lambda_{\mathrm{het}}}$. Then, on E_T^{abs},*

$$\|\sin\Theta(\mathcal{V}_{K,r_{\mathrm{pop}}}, \widehat{\mathcal{V}}_{K,r_{\mathrm{pop}}})\|_{\mathrm{op}} \le \frac{\eta_n^T(\delta)}{\sqrt{\lambda_{\mathrm{het}}} - \eta_n^T(\delta)}.$$

Proof. See Section C.3 for a detailed proof. □

Under the half-gap condition $\eta_n^T(\delta) \le \sqrt{\lambda_{\mathrm{het}}}/2$, the bound simplifies to $2\eta_n^T(\delta)/\sqrt{\lambda_{\mathrm{het}}}$. Thus stronger heterogeneity tightens the $\sin\Theta$ bound at a fixed whitened-operator perturbation level. The envelope $\eta_n^T(\delta)$ is realized by standard sub-Gaussian concentration (Corollary A.5).

3.3 Capacity Limits of the Embedding Bottleneck

This subsection studies the architectural bottleneck itself. Since JEPA predicts through a d-dimensional representation $x = Az_{\mathrm{ctx}}$, its end-to-end map from the context block to the stacked target embedding must factor through a rank at most d map. This induces an approximation limit even when the context contains richer predictive structure. Recall the second moments Σ_{ZZ}, Σ_{YZ} (Section 2, $\Sigma_{ZZ} \succ 0$), the OLS map $B^{\mathrm{raw}}_{\mathrm{OLS}}$ and the whitened operator T_K from (5). The next result formalizes the irreducible approximation error imposed by the d-dimensional embedding bottleneck.

Proposition 3.9 (Capacity decomposition from the embedding bottleneck)**.** *Let $\sigma_1(T_K) \geq \sigma_2(T_K) \geq \cdots \geq 0$ denote the singular values of T_K. For any $r \leq d$,*

$$\inf_{\mathrm{rank}(B)\leq r} \mathbb{E}\left[\|Y - BZ\|_2^2\right] = \mathbb{E}\left[\|Y - B^{\mathrm{raw}}_{\mathrm{OLS}} Z\|_2^2\right] + \sum_{j>r} \sigma_j(T_K)^2. \tag{10}$$

Proof. See Section C.4 for a detailed proof. □

Here r is the reduced-rank constraint (4), which the architecture fixes at the embedding dimension d. The tail $\sum_{j>d} \sigma_j(T_K)^2$ is then irreducible: no choice of (W, A) can recover predictive content beyond the top-d singular directions of T_K. This sharpens the saturation of Section 3.2, since the embedding dimension d is a ceiling that extra target blocks cannot bypass.

3.4 Post-Saturation Gains via Eigenvalue Balancing

By Corollary 3.5, the predictive rank $r_{\mathrm{pop}}(K)$ eventually saturates, so extra targets no longer enlarge the predictive subspace. Its finite-sample recovery, however, still depends on the spectral conditioning of T_K, which we study next. Given $T_K = B^{\mathrm{raw}}_{\mathrm{OLS}} \Sigma_{ZZ}^{1/2}$ (Proposition 3.9) and $H_K \coloneqq T_K^\top T_K$, define

$$\kappa_K^2 \coloneqq \frac{\lambda_d(H_K)}{\lambda_1(H_K)} = \left(\frac{\sigma_d(T_K)}{\sigma_1(T_K)}\right)^2 \in (0, 1]. \tag{11}$$

We focus on the saturated rank-d regime over a finite target window where $r_{\mathrm{pop}}(K) = \mathrm{rank}(T_K) = d$: the predictive rank has reached the architectural ceiling, and κ_K quantifies this conditioning. To separate conditioning from estimator noise, define the relative perturbation event

$$E_T(K; \rho_n) \coloneqq \left\{ \|\widehat{T}_K - T_K\|_{\mathrm{op}} \leq \rho_n(\delta) \|T_K\|_{\mathrm{op}} \right\}. \tag{12}$$

For any $r \leq d$, let $\mathcal{V}_{K,r}$, $\widehat{\mathcal{V}}_{K,r}$ denote the top-r right singular subspaces of T_K and $\widehat{T}_K$, respectively. To translate conditioning-driven subspace stability into excess risk, we establish a Lipschitz relation between the two. For a predictive subspace $\mathcal{V} \in \mathrm{Gr}(r, D_x)$, define

$$\mathcal{R}_K(\mathcal{V}) \coloneqq \frac{1}{K} \inf_{W,A:\, \mathrm{row}(WA) \subseteq \mathcal{V}} \mathbb{E}\left[\|Y - WAZ\|^2\right] \tag{13}$$

as the optimal per-target risk among predictors using $\mathcal{V}$. The factor $1/K$ removes trivial growth in target dimension, so variation in $\mathcal{R}_K$ reflects predictive quality rather than aggregation.

Lemma 3.10 (Subspace-Lipschitz excess risk)**.** *For a fixed target count K and rank $r \leq d$, let $\mathcal{R}^\star_K(r) \coloneqq \mathcal{R}_K(\mathcal{V}_{K,r})$. There exists a constant $L > 0$ (one may take $L = r\,\sigma_1(T_K)^2/K$) such that*

$$\mathcal{R}_K(\widehat{\mathcal{V}}_{K,r}) - \mathcal{R}^\star_K(r) \leq L \|\sin\Theta(\mathcal{V}_{K,r}, \widehat{\mathcal{V}}_{K,r})\|_{\mathrm{op}}.$$

Proof. See Section C.7 for a detailed proof. □

Based on Lemma 3.10, we obtain a risk decomposition of the per-target risk into a population floor and a finite-sample excess term, with the derivation deferred to Section A.3.

Proposition 3.11 (Post-saturation risk decomposition)**.** *Fix $K \in \{K^\star, \ldots, K_{\max}\}$ with $r_\star(K) = \mathrm{rank}(T_K) = d$ and $\kappa_K^2 \geq \kappa_0^2 + \gamma(K - K^\star)$ for some $\kappa_0, \gamma > 0$. Suppose $\mathbb{P}(E_T(K; \rho_n)) \geq 1 - \delta$ and the half-gap condition (24) holds. Then, under Lemma 3.10, on $E_T(K; \rho_n)$,*

$$\mathcal{R}_K(\widehat{\mathcal{V}}_{K,d}) \leq \underbrace{\tfrac{1}{K}\Big(\mathbb{E}\left[\|Y - B^{\mathrm{raw}}_{\mathrm{OLS}} Z\|^2\right] + \sum\nolimits_{j>d} \sigma_j(T_K)^2\Big)}_{\mathcal{R}^\star_K(d),\ \textit{population rank-}d\ \textit{risk}} + \underbrace{\frac{2L\rho_n(\delta)}{\sqrt{\kappa_0^2 + \gamma(K - K^\star)}}}_{\textit{excess risk}}. \tag{14}$$

Proof. See Section C.5 for a detailed proof. □

The excess-risk term in (14) is governed by two finite-sample effects: shrinkage in n at fixed K (Remark 3.12), and post-saturation conditioning gains in K at fixed n (Remark 3.13).

Remark 3.12 (Sample complexity). The risk bound is controlled by the relative perturbation level $\rho_n(\delta)$. If a task-specific estimator analysis gives $\rho_n(\delta) = \widetilde{O}(\sqrt{r_{\mathrm{rel}}/n})$ over the finite comparison window, then $\mathcal{R}_K(\widehat{\mathcal{V}}_{K,d}) - \mathcal{R}^\star_K(d) \le \varepsilon$ is obtained at $n = \widetilde{O}\left(L^2 r_{\mathrm{rel}}/\left((\kappa_0^2 + \gamma(K - K^\star))\varepsilon^2\right)\right)$.

Remark 3.13 (Second saturation). Corollary 3.5 establishes a population saturation of $r_{\mathrm{pop}}(K)$. In the regime $r_{\mathrm{pop}}(K) = d$, Proposition 3.11 reveals a second finite-sample saturation: adding blocks keeps balancing the eigenvalue-conditioning of H_K and improving recovery until κ_K^2 reaches its ceiling of 1.

3.5 Risk Reduction in Multi-Target JEPA

The previous subsections identify two mechanisms: heterogeneity of $\{M_k\}_{k=1}^K$ controls the predictive rank and improves finite-sample identifiability (Propositions 3.4 and 3.8); in the post-saturation regime, recovery improves by balancing the spectrum of $T_K \coloneqq \Sigma_{YZ}\Sigma_{ZZ}^{-1/2}$ (Proposition 3.11). The next theorem unifies them. Let $\mathcal{V}_{K,r}, \widehat{\mathcal{V}}_{K,r}$ be the top-r right singular subspaces of T_K and $\widehat{T}_K$.

For fixed K and $r \le d$, define the absolute gap $\Delta^{\mathrm{abs}}_{K,r} \coloneqq \sigma_r(T_K) - \sigma_{r+1}(T_K)$, with the convention $\sigma_{r+1}(T_K) = 0$ when $r = \mathrm{rank}(T_K)$, and the relative gap $\bar{\Delta}_{K,r} \coloneqq \Delta^{\mathrm{abs}}_{K,r}/\|T_K\|_{\mathrm{op}}$. Proposition 3.11 requires a half-gap condition (24); for the general rank-r decomposition, we only require the relative perturbation level to be smaller than the corresponding relative Wedin gap:

Assumption 3.14 (Gap-compatible relative perturbation)**.** For the fixed target count K and rank $r \le d$, the event $E_T(K;\rho_n)$ in (12) satisfies $\mathbb{P}(E_T(K;\rho_n)) \ge 1 - \delta$ for some $\rho_n(\delta) < \bar{\Delta}_{K,r}$.

Theorem 3.15 (Unified risk decomposition)**.** *Under Assumption 3.14 and Lemma 3.10, on $E_T(K;\rho_n)$,*

$$\mathcal{R}_K(\widehat{\mathcal{V}}_{K,r}) \le \underbrace{\frac{1}{K}\left(\mathbb{E}\left[\|Y - B^{\mathrm{raw}}_{\mathrm{OLS}}Z\|^2\right] + \sum_{j>r}\sigma_j(T_K)^2\right)}_{\mathcal{R}^\star_K(r),\ \textit{population rank-r risk}} + \underbrace{L\frac{\rho_n(\delta)}{\bar{\Delta}_{K,r} - \rho_n(\delta)}}_{\textit{excess-risk}}. \tag{15}$$

Proof. See Section C.6 for a detailed proof. □

Remark 3.16 (Risk-reduction levers). The bound (15) isolates two controllable levers given fixed targets and context. *Capacity*: increasing K can increase $r_{\mathrm{pop}}(K) = \mathrm{rank}(T_K)$, but the architecture keeps at most d directions. If $r_{\mathrm{pop}}(K) > d$, the residual tail $\frac{1}{K}\sum_{j>d}\sigma_j(T_K)^2$ remains. *Conditioning*: finite-sample recovery depends on the relative singular gap $\bar{\Delta}_{K,r}$, which at $r_{\mathrm{pop}}(K) = d$ reduces to $\kappa_K = \sigma_d(T_K)/\sigma_1(T_K)$. The remaining task-predictability floor $\frac{1}{K}\mathbb{E}\|Y - B^{\mathrm{raw}}_{\mathrm{OLS}}Z\|^2$ is irreducible and tracked by $r_{\mathrm{pop}}(K)$ (Definition 3.3).

4 Predictive Isotropy

Theorem 3.15 decomposes the empirical risk of JEPA. In this section, we bridge its risk minimization to a predictive isotropy condition (Section 4.1), and then show that this condition is gauge-invariant, thus subsuming and unifying existing encoder-isotropy designs into a regularization (Section 4.2).

4.1 Unique Risk Minimizer

Recall the stacked context $Z \in \mathbb{R}^{D_x}$ from Section 3.3, the linear context encoder A from Assumption 2.3, and the whitened end-to-end map T_K with $H_K \coloneqq T_K^\top T_K$ from Proposition 3.9.

The object of interest is the JEPA prediction $P_K \coloneqq T_K\widetilde{Z}$, i.e., $P_K = B^{\mathrm{raw}}_{\mathrm{OLS}}Z$ written in whitened coordinates. The rank-d prediction WAZ approximates P_K under the embedding bottleneck (10). Whereas LeJEPA's SIGReg regularizes the law of the embedding AZ toward an isotropic Gaussian prior [5], we target the prediction-aware analogue on P_K.

Lemma 4.1 (Predictive isotropy)**.** *Let $H_K = T_K^\top T_K$. Assuming $\Sigma_{ZZ} \succ 0$, $\mathbb{E}[\widetilde{Z}] = 0$, and $\mathrm{rank}(T_K) = d$, the JEPA prediction $P_K = T_K\widetilde{Z}$ has isotropic covariance on its rank-d predictive range if and only if*

$$\lambda_1(H_K) = \lambda_2(H_K) = \cdots = \lambda_d(H_K). \tag{16}$$

Proof. See Section C.8 for a detailed proof. □

We show predictive isotropy (Lemma 4.1) is the unique minimizer of the Wedin excess-risk bound.

Theorem 4.2 (Predictive isotropy minimizes excess-risk bound)**.** *Fix $\tau > 0$, a relative perturbation level $\rho_n(\delta)$, and consider operators T_K satisfying $\operatorname{rank}(T_K) = d$, $\operatorname{tr}(H_K) = \|T_K\|_F^2 = \tau$ and $\rho_n(\delta) < \kappa_K$. Among all such operators, the isotropic spectrum $\lambda_1(H_K) = \cdots = \lambda_d(H_K) = \tau/d$ uniquely minimizes the Wedin excess-risk upper bound $\rho_n(\delta)/(\kappa_K - \rho_n(\delta))$ in Theorem 3.15.*

Proof. See Section C.9 for a detailed proof. □

The recovery-bound optimum in Theorem 4.2 is an instance of a broader principle, see Section A.4.

4.2 Regularization

Lemma 4.1 admits a distributional reading that places the present subsection one step further down the JEPA pipeline than LeJEPA's SIGReg. Recall that for a measure μ and a map f, the pushforward $f_{\#}\mu$ denotes the law of $f(X)$ when $X \sim \mu$. SIGReg regularizes the embedding pushforward $\mathcal{L}(AZ) = A_{\#}\mathcal{L}(Z)$ toward an isotropic Gaussian prior [5]; the prediction-aware analogue is the predictive pushforward

$$\mathcal{L}(P_K) = (T_K)_{\#}\mathcal{L}(\widetilde{Z}). \tag{17}$$

Remark 4.3 (Relation to encoder isotropic regularization). Covariance isotropy of the predictive pushforward $\mathcal{L}(P_K)$ (17) on its rank-d range is exactly the eigenvalue-balancing condition of $H_K = T_K^\top T_K$ (Lemma 4.1). Full Gaussianity is a stronger maximum-entropy completion (Lemma A.10).

Within Lemma 4.1's condition, the encoder-predictor factorization remains a structural choice. We specify one that bridges the predictive isotropy with the embedding isotropy of canonical JEPA paradigms [5]. Recall the SVD of the whitened end-to-end map, $T_K = U_K S_K V_K^\top$, and let $V_K^\top \widetilde{Z} \in \mathbb{R}^d$ be the *SVD coordinate* of the whitened context.

Lemma 4.4 (Gauge invariance of the rank-d optimum)**.** *Let a gauge $G : \mathbb{R}^d \to \mathbb{R}^d$ be any measurable map with an a.e. inverse G^{-1} on the support. Define an encoder-predictor factorization (Λ_G, F_G)*

$$\Lambda_G \coloneqq G \circ V_K^\top \Sigma_{ZZ}^{-1/2}, \qquad F_G \coloneqq U_K S_K \circ G^{-1}.$$

Then $F_G \circ \Lambda_G$ realizes the population rank-d prediction P_K for every choice of G.

By Lemma 4.4, predictive isotropy (Lemma 4.1) is gauge-invariant, so it constrains T_K without committing to any embedding distribution. The gauge G is therefore free to be chosen for downstream purposes. In particular, it can be selected to realize either of the two canonical encoder regularization paradigms in JEPA: covariance isotropy (whitening [16] or VICReg-style [6]) or Gaussian isotropy (LeJEPA [5], SIGReg-style).

Theorem 4.5 (Subsumption of covariance isotropy)**.** *For any orthogonal gauge $G : \mathbb{R}^d \to \mathbb{R}^d$, the encoder-predictor factorization (Λ_G, F_G) in Lemma 4.4 induces an isotropic covariance embedding $\operatorname{Cov}(\Lambda_G(Z)) = I_d$.*

Proof. See Section C.10 for a detailed proof. □

Theorem 4.5 realizes covariance isotropy with a linear gauge. We next show that a nonlinear gauge can realize the strictly stronger property of isotropic-Gaussian embedding under a mild assumption.

Theorem 4.6 (Subsumption of Gaussian isotropy)**.** *Assume $V_K^\top \widetilde{Z}$ has absolutely continuous law on $\mathbb{R}^d$. Then there exists a gauge $G^\star : \mathbb{R}^d \to \mathbb{R}^d$ such that the encoder-predictor factorization $(\Lambda_{G^\star}, F_{G^\star})$ in Lemma 4.4 induces an isotropic Gaussian embedding $\Lambda_{G^\star}(Z) \sim \mathcal{N}(0, I_d)$.*

Proof Sketch. Existence of $G^\star$ follows from Brenier's theorem [17], since the law of $V_K^\top \widetilde{Z}$ admits a density on $\mathbb{R}^d$ by hypothesis and $\mathcal{N}(0, I_d)$ admits a density on $\mathbb{R}^d$. Using an a.e. inverse on the support, the encoder-predictor pair $(\Lambda_{G^\star}, F_{G^\star})$ then satisfies the conclusion by construction (Lemma 4.4). □

When the context is Gaussian, the gauge in Theorem 4.6 collapses to the identity.

Corollary 4.7 (Linear-Gaussian special case)**.** *If the context follow a Gaussian distribution $Z \sim \mathcal{N}(0, \Sigma_{ZZ})$, the encoder-predictor factorization reduces to the linear pair*

$$\Lambda_I = V_K^\top \Sigma_{ZZ}^{-1/2}, \qquad F_I = U_K S_K.$$

Proof. See Section C.11 for a detailed proof. □

Subsumption in Theorems 4.5 and 4.6 means the converse fails: encoder isotropy alone does not imply predictive isotropy, since it is compatible with arbitrary predictor conditioning $\kappa_K^2 \in (0, 1]$. Moreover, by Lemma 4.4, every choice of G realizes the same prediction $P_K = T_K \widetilde{Z}$ with the same optimality condition in Theorem 4.2, so selecting a gauge in Theorems 4.5 and 4.6 costs nothing against the excess-risk bound.

We now turn this principle into a regularizer. The excess-risk bound in Theorem 3.15 motivates spectral isotropy of the gauge-free predictive object H_K. The subsumption results in Theorems 4.5 and 4.6 further show that, under such regularization, encoder isotropy need not be imposed separately, nor must a gauge in Lemma 4.4 be chosen manually.

Maximum-entropy regularization induces isotropic-Gaussian embedding. The excess-risk bound prescribes only the spectral condition (16) on H_K, equivalently covariance isotropy of P_K on its rank-d predictive range. This places candidate regularizers on a two-axis frontier: *fidelity*, whether $\mathcal{R} = 0$ implies (16), and *commitment*, how much distributional structure $\mathcal{R} = 0$ enforces beyond it. The least-committing faithful choice is the maximum-entropy law satisfying predictive isotropy. At trace τ, Lemma A.10 shows that this law is uniquely $\mathcal{N}(0, \frac{\tau}{d} I_d)$ on the intrinsic predictive range.

Therefore, we propose a maximum-entropy regularizer for the intrinsic predicted output $\widehat{P}_K^{\mathrm{int}} := \widehat{U}_K^\top \widehat{W} \widehat{A} Z \in \mathbb{R}^d$, where $\widehat{U}_K \in \mathbb{R}^{Kd \times d}$ spans the learned rank-d predictive range. By the Cramér-Wold theorem [18], in its hyperspherical form [5, Lemma 3], the target law $\widehat{P}_K^{\mathrm{int}} \sim \mathcal{N}(0, I_d)$ is equivalent to $u^\top \widehat{P}_K^{\mathrm{int}} \sim \mathcal{N}(0, 1)$ for every $u \in \mathbb{S}^{d-1}$. We approximate this all-direction condition by sampling $|\mathbb{A}|$ random unit vectors $\mathbb{A} \subset \mathbb{S}^{d-1}$, projecting the predictor output onto each direction, and averaging the standard-normal Gaussianity statistics T (Epps-Pulley [19]):

$$\mathcal{R}_{\mathrm{PI}}(\widehat{P}_K^{\mathrm{int}}; \mathbb{A}) := \frac{1}{|\mathbb{A}|} \sum_{u \in \mathbb{A}} T\left(\{u^\top \widehat{p}_n\}_{n=1}^N\right), \tag{18}$$

where $\widehat{p}_n := \widehat{U}_K^\top \widehat{W} \widehat{A} z_n$ is the intrinsic predicted output on the n-th sample.

By Lemma 4.1, the covariance part of $\mathcal{N}(0, I_d)$ enforces predictive isotropy on the nonzero spectrum of H_K, which minimizes the excess-risk bound (Theorem 4.2). The Gaussian part is its maximum-entropy completion. (18) is therefore the predictive analogue of LeJEPA's SIGReg [5], constraining $\mathcal{L}(\widehat{P}_K^{\mathrm{int}})$ rather than $\mathcal{L}(AZ)$. Practically, (18) implements this principle through a sample-level Gaussianity surrogate on $\widehat{P}_K^{\mathrm{int}}$. Together with Theorems 4.5 and 4.6, it justifies and unifies the two encoder regularization paradigms in canonical JEPA [16, 6, 5] as factorized instances of the same predictive-isotropy principle.

5 Experimental Studies

5.1 Synthetic Experiment

Setup. We first use frozen-teacher linear experiments with known population operators to isolate the mechanisms in the theory from nonlinear optimization effects. Full setup details are in Section D.1.

Models and baselines. The learned model is the finite-sample RRR solution. Rank d instantiates the JEPA bottleneck. Population operators and unconstrained OLS serve as baselines.

Metrics. Risk panels use the per-target normalization in (13). Diagnostics match the quantities in Theorem 3.15 and the rank, heterogeneity, and conditioning results. Definitions are in Section D.1.

Results. The appendix figures Figure 3, Figures 4 to 6 validate the reduction, pre-saturation, and bottleneck claims. The factorized linear predictor realizes the RRR solution; complementary target stacks increase predictive rank, larger λ_{het} improves finite-sample identifiability, and the rank-d bottleneck induces the spectral-tail floor predicted by Propositions 3.4, 3.8 and 3.9. Figure 7 independently checks Theorem 3.15: with $r < r_{\mathrm{pop}}(K)$, the population spectral tail is nonzero, the retained-gap condition is non-vacuous, and the calibrated excess term bounds the observed finite-sample risk. The main synthetic diagnostic is Figure 1. It tests Proposition 3.11 when $r_{\mathrm{pop}}(K) = d$, where additional target blocks cannot create new rank. The experiment allocates new blocks to weak predictive directions. As $\kappa_K^2 = \lambda_d(H_K)/\lambda_1(H_K)$ increases, the observed excess risk and per-target risk decrease. Figure 8 tests Lemma 4.1 and Theorem 4.2: at fixed rank and trace, flatter spectra of $H_K = T_K^\top T_K$ improve recovery and reduce the balanced-reference-stack excess risk. The gauge-factorization diagnostic in Figure 10 checks that the encoder-isotropy claims in Section 4.2 are factorization effects rather than new end-to-end mapping constraints.

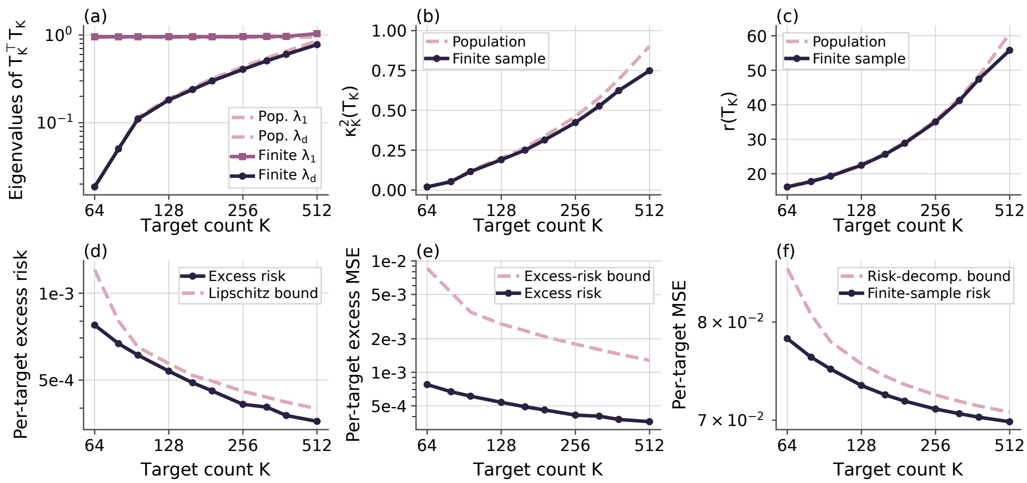

Figure 1: **Post-saturation conditioning.** Given $r_{\mathrm{pop}}(K) = d$, allocating targets to weak directions increases κ_K^2 and reduces excess risk and per-target risk. Panel (d) plots the calibrated Lipschitz bound $\widehat{L}\|\sin\Theta_T\|_{\mathrm{op}}$.

Table 1: **Real-image regularizer diagnostics.** MSE and gap columns are reported in 10^{-2} units, and RMSE is reported as a percentage of the normalized $[0, 1]$ pixel range. Gap is test MSE minus train MSE. Foreground MSE is reported only for MNIST, where background pixels dominate the right-half target.

Task	Objective	Test MSE	RMSE (%)	Test/Train Gap	FG MSE	Time
8×8 digits	No reg.	9.23±0.17	30.4±0.3	8.54±0.17	-	9.03±0.20 s (1.00×)
8×8 digits	Enc. reg.	7.86±0.13	28.0±0.2	6.29±0.14	-	12.55±0.11 s (1.39×)
8×8 digits	Pred. reg.	**6.00±0.03**	**24.5±0.1**	**2.00±0.07**	-	12.23±0.04 s (1.35×)
8×8 digits	Pred.+Enc.	6.09±0.04	24.7±0.1	2.27±0.08	-	15.80±0.13 s (1.75×)
MNIST CNN	No reg.	3.68±0.02	19.19±0.04	2.81±0.05	12.90±0.07	69.52±4.23 s (1.00×)
MNIST CNN	Enc. reg.	3.65±0.02	19.11±0.05	2.68±0.05	12.75±0.07	55.39±0.13 s (0.80×)
MNIST CNN	Pred. reg.	**3.43±0.02**	**18.53±0.04**	2.13±0.06	**11.92±0.06**	53.98±0.07 s (0.78×)
MNIST CNN	Pred.+Enc.	3.47±0.01	18.63±0.04	**2.12±0.05**	12.06±0.05	39.08±4.30 s (0.56×)

5.2 Realistic Experiment

Setup. We test the practical regularizer (18) in learned nonlinear image inpainting tasks. In both tasks the context is the left half of a digit image and the target is the right half. Figure 2 shows the task.

Data. The small-data diagnostic uses the `sklearn` 8×8 digit dataset. The scaled diagnostic uses standard 28×28 MNIST. All pixels are normalized to $[0, 1]$.

Models. The small-data task uses an MLP encoder-predictor, and MNIST uses a convolutional encoder-decoder. The prediction-side regularizer is applied to the predictor representation as in (18); architecture and optimizer details are in Section D.2.

Baselines. We compare four objectives: no regularization, Encoder Gaussianity, prediction-side Gaussianity, and their combination. Encoder Gaussianity is the standard representation-side control. Prediction-side Gaussianity is our theory-motivated regularizer in (18), and the combined objective is an ablation testing whether Encoder Gaussianity improves on prediction-side Gaussianity.

Metrics. We report target-half pixel MSE and RMSE. For MNIST we also report foreground MSE, computed on non-background right-half pixels as most right-half pixels are background. Gaussianity and effective-rank diagnostics are reported in Figures 11 and 12; prediction risk is the primary outcome.

Results. Figure 2 visualizes the digit-half prediction task, and Table 1 reports the regularizer diagnostics. Additional completion examples are shown in Figures 13 and 14. On the low-label 8×8 digit task, prediction-side regularization reduces test MSE from 9.23×10^{-2} to 6.00×10^{-2} and train–test gap from 8.54×10^{-2} to 2.00×10^{-2}. On the scaled MNIST task, prediction-side regularization improves test MSE from 3.68×10^{-2} to 3.43×10^{-2} and foreground MSE from 1.29×10^{-1} to 1.19×10^{-1}. The combined objective adds little, consistent with the subsumption view in Theorems 4.5 and 4.6.

6 Concluding Remarks

We give a finite-sample ERM analysis of JEPA and identify an optimality condition. Casting JEPA as reduced-rank regression (Proposition 3.1), we decompose its risk into population and excess terms (Theorem 3.15). The excess term is uniquely minimized by a flat-spectrum condition termed

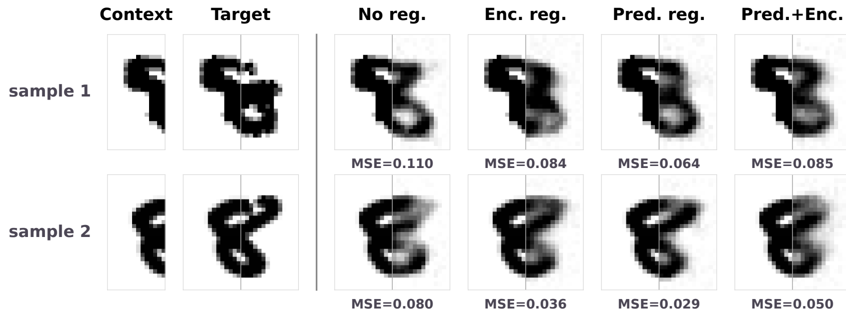

Figure 2: **MNIST half-completions.** Rows are held-out samples; columns show context, ground truth, and candidate completions. Cell labels report per-sample target-half pixel MSE.

predictive isotropy (Theorem 4.2). The population term is controlled by the optimal predictor's rank, which grows with target heterogeneity and saturates at the embedding bottleneck (Propositions 3.4, 3.8 and 3.9). Because the risk is gauge-invariant, predictive isotropy pins down the end-to-end map but leaves its encoder–predictor factorization free; existing encoder-isotropy recipes each fix a gauge realizing it (Lemma 4.4 and Theorems 4.5 and 4.6). Acting on the map, we derive a predictive-isotropy regularizer (18) that subsumes isotropic-Gaussian embedding without fixing a gauge. We defer discussion with related work to Section B.

References

[1] Yann LeCun et al. A path towards autonomous machine intelligence version 0.9. 2, 2022-06-27. *Open Review*, 62(1):1–62, 2022.

[2] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 15619–15629, 2023.

[3] Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mido Assran, and Nicolas Ballas. V-jepa: Latent video prediction for visual representation learning. 2023.

[4] Adrien Bardes, Jean Ponce, and Yann LeCun. Mc-jepa: A joint-embedding predictive architecture for self-supervised learning of motion and content features. *arXiv preprint arXiv:2307.12698*, 2023.

[5] Randall Balestriero and Yann LeCun. Lejepa: Provable and scalable self-supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544*, 2025.

[6] Adrien Bardes, Jean Ponce, and Yann LeCun. Vicreg: Variance-invariance-covariance regularization for self-supervised learning. *arXiv preprint arXiv:2105.04906*, 2021.

[7] Jean-Bastien Grill, Florian Strub, Florian Altché, Corentin Tallec, Pierre H Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, et al. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems*, 2020.

[8] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9650–9660, 2021.

[9] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pages 1597–1607. PmLR, 2020.

[10] Li Jing, Pascal Vincent, Yann LeCun, and Yuandong Tian. Understanding dimensional collapse in contrastive self-supervised learning. In *International Conference on Learning Representations (ICLR)*, 2022.

[11] Yuandong Tian, Xinlei Chen, and Surya Ganguli. Understanding self-supervised learning dynamics without contrastive pairs. In *International Conference on Machine Learning*, pages 10268–10278. PMLR, 2021.

[12] Sofiane Mansouri et al. Understanding early collapse in predictive world-model pretraining. *OpenReview*, 2026.

[13] Randall Balestriero, Nicolas Ballas, Mike Rabbat, and Yann LeCun. Gaussian embeddings: How jepas secretly learn your data density. *arXiv preprint arXiv:2510.05949*, 2025.

[14] Theodore Wilbur Anderson. Estimating linear restrictions on regression coefficients for multivariate normal distributions. *The Annals of Mathematical Statistics*, pages 327–351, 1951.

[15] Alan Julian Izenman. Reduced-rank regression for the multivariate linear model. *Journal of multivariate analysis*, 5(2):248–264, 1975.

[16] Aleksandr Ermolov, Aliaksandr Siarohin, Enver Sangineto, and Nicu Sebe. Whitening for self-supervised representation learning. In *ICML*, 2021.

[17] Cédric Villani et al. *Optimal transport: old and new*, volume 338. Springer, 2009.

[18] Harald Cramér and Herman Wold. Some theorems on distribution functions. *Journal of the London Mathematical Society*, s1-11(4):290–294, 1936.

[19] T. W. Epps and Lawrence B. Pulley. A test for normality based on the empirical characteristic function. *Biometrika*, 70(3):723–726, 1983.

[20] Gregory C Reinsel and Raja P Velu. *Multivariate reduced-rank regression*. Springer, 1998.

[21] Vladimir Koltchinskii and Karim Lounici. Concentration inequalities and moment bounds for sample covariance operators. *Bernoulli*, pages 110–133, 2017.

[22] Per-Åke Wedin. Perturbation bounds in connection with singular value decomposition. *BIT Numerical Mathematics*, 12(1):99–111, 1972.

[23] A Wayne Roberts and Dale E Varberg. *Convex Functions: Convex Functions*, volume 57. Academic Press, 1974.

[24] Albert W Marshall, Ingram Olkin, and Barry C Arnold. Inequalities: theory of majorization and its applications. 1979.

[25] Nikunj Saunshi, Orestis Plevrakis, Sanjeev Arora, Mikhail Khodak, and Hrishikesh Khandeparkar. A theoretical analysis of contrastive unsupervised representation learning. In *International conference on machine learning*, pages 5628–5637. PMLR, 2019.

[26] Christopher Tosh, Akshay Krishnamurthy, and Daniel Hsu. Contrastive estimation reveals topic posterior information to linear models. *Journal of Machine Learning Research*, 22(281):1–31, 2021.

[27] Jeff Z HaoChen, Colin Wei, Adrien Gaidon, and Tengyu Ma. Provable guarantees for self-supervised deep learning with spectral contrastive loss. *Advances in neural information processing systems*, 34:5000–5011, 2021.

[28] Jason D Lee, Qi Lei, Nikunj Saunshi, and Jiacheng Zhuo. Predicting what you already know helps: Provable self-supervised learning. *Advances in Neural Information Processing Systems*, 34:309–323, 2021.

[29] Yuandong Tian, Lantao Yu, Xinlei Chen, and Surya Ganguli. Understanding self-supervised learning with dual deep networks. *arXiv preprint arXiv:2010.00578*, 2020.

[30] Bobak T Kiani, Randall Balestriero, Yubei Chen, Seth Lloyd, and Yann LeCun. Joint embedding self-supervised learning in the kernel regime. *arXiv preprint arXiv:2209.14884*, 2022.

[31] Vivien Cabannes, Bobak Kiani, Randall Balestriero, Yann LeCun, and Alberto Bietti. The ssl interplay: Augmentations, inductive bias, and generalization. In *International conference on machine learning*, pages 3252–3298. PMLR, 2023.

[32] Etai Littwin, Omid Saremi, Madhu Advani, Vimal Thilak, Preetum Nakkiran, Chen Huang, and Joshua Susskind. How jepa avoids noisy features: The implicit bias of deep linear self distillation networks. *Advances in Neural Information Processing Systems*, 37:91300–91336, 2024.

[33] Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and Stéphane Deny. Barlow twins: Self-supervised learning via redundancy reduction. In *International conference on machine learning*, pages 12310–12320. PMLR, 2021.

[34] Carl Eckart and Gale Young. The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3):211–218, 1936.

Appendix

Impact Statement

By the theoretical nature of this work, we do not anticipate any negative social impact.

LLM Usage Disclosure

We used large language models (LLMs) to aid and polish writing, such as improving clarity, grammar, and conciseness. We also used LLMs for retrieval and discovery, for example, literature-search assistance to identify potential missing related work. In addition, LLM-based coding assistants were used as basic code assistance to help implement the numerical validation experiments (e.g., plotting and data pipelines). All technical content, proofs, experiments, and results are original contributions by the authors.

A Supplementary Theoretical Backgrounds

A.1 Reduced-Rank Regression

Given paired random vectors $x \in \mathbb{R}^p$ and $y \in \mathbb{R}^q$, the population squared-loss regression problem

$$\min_{B \in \mathbb{R}^{q \times p}} \mathbb{E}\big[\|y - Bx\|^2\big] \tag{19}$$

admits the closed-form ordinary least squares (OLS) minimizer $B_{\mathrm{OLS}} := \Sigma_{yx}\Sigma_{xx}^{-1}$ when $\Sigma_{xx} \succ 0$. Reduced-rank regression (RRR) [15, 20] restricts the minimization in (19) to $\mathrm{rank}(B) \leq r$.

Definition A.1 (Reduced-rank regression problem). Given paired $x \in \mathbb{R}^p$, $y \in \mathbb{R}^q$ as predictor and response random vectors, for a prescribed rank $r \leq \min(p, q)$, the reduced-rank regression is

$$B_{\mathrm{RRR}} \in \arg\min_{B \in \mathbb{R}^{q \times p}} \mathbb{E}\big[\|y - Bx\|^2\big] \quad \text{subject to} \quad \mathrm{rank}(B) \leq r. \tag{20}$$

The constraint $\mathrm{rank}(B) \leq r$ means the linear predictor factors through an r-dimensional latent subspace. The following lemma gives a closed-form solution.

Lemma A.2 (RRR solution, Theorem 2.2 of [20]). *Let $V_r \in \mathbb{R}^{p \times r}$ collect eigenvectors corresponding to the top r eigenvalues of $\Sigma_{XX}^{1/2} B_{\mathrm{OLS}}^\top B_{\mathrm{OLS}} \Sigma_{XX}^{1/2}$. A solution to (20) is*

$$B_{\mathrm{RRR}} = B_{\mathrm{OLS}} \Sigma_{XX}^{1/2} V_r V_r^\top \Sigma_{XX}^{-1/2}. \tag{21}$$

Equivalently, RRR solution is obtained by taking the best rank-r approximation of $B_{\mathrm{OLS}}\Sigma_{XX}^{1/2}$ and then unwhitening by $\Sigma_{XX}^{-1/2}$. Informally, this means that RRR retains the r predictor-space directions that are most useful for linearly predicting y.

Remark A.3 (Connection to CCA and PCA). Reduced-rank regression is closely related to canonical correlation analysis (CCA): after appropriate whitening, the rank-r solution retains the leading canonical predictive directions linking the predictor space to the response space. As a special case, when $y = x$, the corresponding reduced-rank self-prediction problem recovers the principal subspace of Σ_{XX}, linking RRR to PCA.

A.2 Useful Lemmas

Lemma A.4 (Joint-covariance control of cross-covariance noise [21]). *Let $(Z_i, Y_i)_{i=1}^n$ be i.i.d. samples with $Z \in \mathbb{R}^{D_x}$ and $Y \in \mathbb{R}^{Kd}$, and assume $U := (Y^\top, Z^\top)^\top$ is sub-Gaussian with second moment $M := \mathbb{E}[UU^\top]$. Let $\widehat{M} := n^{-1}\sum_{i=1}^n U_i U_i^\top$ and $\widehat{\Sigma}_{YZ} := n^{-1}\sum_{i=1}^n Y_i Z_i^\top$. Then $\widehat{\Sigma}_{YZ} - \Sigma_{YZ}$ is an off-diagonal block of $\widehat{M} - M$. Consequently, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\|\widehat{\Sigma}_{YZ} - \Sigma_{YZ}\|_{\mathrm{op}} \leq C\|M\|_{\mathrm{op}} \left(\sqrt{\frac{r(M) + \log(1/\delta)}{n}} + \frac{r(M) + \log(1/\delta)}{n} \right),$$

where $C > 0$ is an absolute constant and $r(M) := \mathrm{tr}(M)/\|M\|_{\mathrm{op}}$ is the effective rank of the joint second-moment matrix.

Proof. Write

$$U_i := \begin{pmatrix} Y_i \\ Z_i \end{pmatrix}, \quad M := \mathbb{E}[UU^\top], \quad \widehat{M} := \frac{1}{n}\sum_{i=1}^n U_i U_i^\top.$$

Then

$$M = \begin{pmatrix} \mathbb{E}[YY^\top] & \mathbb{E}[YZ^\top] \\ \mathbb{E}[ZY^\top] & \mathbb{E}[ZZ^\top] \end{pmatrix} = \begin{pmatrix} \Sigma_{YY} & \Sigma_{YZ} \\ \Sigma_{ZY} & \Sigma_{ZZ} \end{pmatrix},$$

and hence

$$\widehat{M} - M = \begin{pmatrix} \widehat{\Sigma}_{YY} - \Sigma_{YY} & \widehat{\Sigma}_{YZ} - \Sigma_{YZ} \\ \widehat{\Sigma}_{ZY} - \Sigma_{ZY} & \widehat{\Sigma}_{ZZ} - \Sigma_{ZZ} \end{pmatrix}.$$

Let $E := \widehat{\Sigma}_{YZ} - \Sigma_{YZ}$. For any unit vectors a, b, define $u = (a^\top, 0^\top)^\top$ and $v = (0^\top, b^\top)^\top$. Then $a^\top E b = u^\top(\widehat{M} - M)v$, so

$$\|E\|_{\mathrm{op}} \le \|\widehat{M} - M\|_{\mathrm{op}}.$$

Applying the effective-rank sample-covariance concentration bound for sub-Gaussian vectors to $\widehat{M} - M$ gives

$$\|\widehat{M} - M\|_{\mathrm{op}} \le C\|M\|_{\mathrm{op}} \left(\sqrt{\frac{r(M) + \log(1/\delta)}{n}} + \frac{r(M) + \log(1/\delta)}{n} \right),$$

which proves the claim. □

Corollary A.5 (Whitened-operator envelope)**.** *Let* $\Sigma_{ZZ} \succ 0$ *and* $\widehat{T}_K = \widehat{\Sigma}_{YZ}\Sigma_{ZZ}^{-1/2}$. *With probability at least* $1 - \delta$,

$$\|\widehat{T}_K - T_K\|_{\mathrm{op}} \le \eta_n^T(\delta) := \frac{\eta_n^{YZ}(\delta)}{\sqrt{\lambda_{\min}(\Sigma_{ZZ})}},$$

where $\eta_n^{YZ}(\delta)$ *denotes the right-hand side of Lemma A.4.*

Proof. Since $T_K = \Sigma_{YZ}\Sigma_{ZZ}^{-1/2}$, we have $\widehat{T}_K - T_K = (\widehat{\Sigma}_{YZ} - \Sigma_{YZ})\Sigma_{ZZ}^{-1/2}$, so submultiplicativity and $\|\Sigma_{ZZ}^{-1/2}\|_{\mathrm{op}} = \lambda_{\min}(\Sigma_{ZZ})^{-1/2}$ give

$$\|\widehat{T}_K - T_K\|_{\mathrm{op}} \le \|\widehat{\Sigma}_{YZ} - \Sigma_{YZ}\|_{\mathrm{op}}\, \|\Sigma_{ZZ}^{-1/2}\|_{\mathrm{op}} = \frac{\|\widehat{\Sigma}_{YZ} - \Sigma_{YZ}\|_{\mathrm{op}}}{\sqrt{\lambda_{\min}(\Sigma_{ZZ})}}.$$

Lemma A.4 bounds the numerator by $\eta_n^{YZ}(\delta)$ with probability at least $1 - \delta$. □

Lemma A.6 (Wedin-type subspace perturbation [22])**.** *Let* $M \in \mathbb{R}^{m \times p}$, *let* $r \le \min(m, p)$, *and let* $\widehat{M}$ *be a perturbation with* $\|\widehat{M} - M\|_{\mathrm{op}} \le \epsilon$. *Denote by* $\mathcal{V}_r, \widehat{\mathcal{V}}_r$ *the top-*r *right singular subspaces of* M *and* $\widehat{M}$, *and define the* Wedin gap *at level* r *as* $g_r(M) := \sigma_r(M) - \sigma_{r+1}(M)$, *with the convention* $\sigma_{r+1}(M) := 0$ *when* $\mathrm{rank}(M) \le r$. *Provided* $\epsilon < g_r(M)$,

$$\|\sin\Theta(\mathcal{V}_r, \widehat{\mathcal{V}}_r)\|_{\mathrm{op}} \;\le\; \frac{\epsilon}{g_r(M) - \epsilon}.$$

Proof. Apply Wedin's $\sin\Theta$ theorem [22] to M and $\widehat{M}$ with spectral gap $g_r(M)$. □

A.3 Intermediate Derivation of Post-Saturation Recovery

This appendix collects the subspace-recovery results that underpin Proposition 3.11 in the body. The derivation proceeds in two steps. First, a Wedin-type perturbation bound translates the deterministic conditioning hypotheses and the relative perturbation event (12) into a finite-sample $\sin\Theta$ bound on the top-d right singular subspace of T_K (Proposition A.7). Second, a half-gap condition exposes the explicit dependence on the conditioning parameter (Corollary A.8). The body result is then assembled in Proposition 3.11 by composing this $\sin\Theta$ bound with the subspace-Lipschitz Lemma 3.10.

Step 1: Wedin sin-Θ bound.

Proposition A.7 (Post-saturation recovery under relative perturbation)**.** *Fix $K \in \{K^\star, \dots, K_{\max}\}$ with $r_{\mathrm{pop}}(K) = \mathrm{rank}(T_K) = d$ and $\kappa_K^2 \geq \kappa_0^2 + \gamma(K - K^\star)$. For any $\delta \in (0,1)$, suppose $\mathbb{P}(E_T(K;\rho_n)) \geq 1-\delta$. Then, on $E_T(K;\rho_n)$ (12),*

$$\|\sin\Theta(\mathcal{V}_{K,d}, \widehat{\mathcal{V}}_{K,d})\|_{\mathrm{op}} \leq \frac{\rho_n(\delta)}{\sqrt{\kappa_0^2 + \gamma(K-K^\star)} - \rho_n(\delta)}, \tag{22}$$

provided the denominator is positive.

Proof. Since $\mathrm{rank}(T_K) = d$, the Wedin gap $g_d(T_K)$ reduces to $\sigma_d(T_K)$. On $E_T(K;\rho_n)$ (12),

$$\|\sin\Theta(\mathcal{V}_{K,d}, \widehat{\mathcal{V}}_{K,d})\|_{\mathrm{op}} \leq \frac{\rho_n(\delta)\|T_K\|_{\mathrm{op}}}{\sigma_d(T_K) - \rho_n(\delta)\|T_K\|_{\mathrm{op}}}, \tag{23}$$

by Lemma A.6. Since $\sigma_d(T_K) = \kappa_K \|T_K\|_{\mathrm{op}}$, division by $\|T_K\|_{\mathrm{op}} > 0$ gives

$$\|\sin\Theta(\mathcal{V}_{K,d}, \widehat{\mathcal{V}}_{K,d})\|_{\mathrm{op}} \leq \frac{\rho_n(\delta)}{\kappa_K - \rho_n(\delta)}.$$

Using $\kappa_K \geq \sqrt{\kappa_0^2 + \gamma(K-K^\star)}$ yields (22). □

Step 2: Explicit K-rate via the half-gap condition. The bound (22) couples the relative perturbation envelope and the deterministic floor in a single ratio. To expose the conditioning dependence, we impose a *half-gap condition* requiring the relative perturbation to be at most half the relative singular-value gap:

$$\rho_n(\delta) \leq \frac{1}{2}\sqrt{\kappa_0^2 + \gamma(K-K^\star)}. \tag{24}$$

This half-gap condition is verified empirically across the studied K-range in Figure 1.

Corollary A.8 (Finite-window post-saturation rate)**.** *Fix $K \in \{K^\star, \dots, K_{\max}\}$ with $r_{\mathrm{pop}}(K) = \mathrm{rank}(T_K) = d$ and $\kappa_K^2 \geq \kappa_0^2 + \gamma(K-K^\star)$. For any $\delta \in (0,1)$, suppose $\mathbb{P}(E_T(K;\rho_n)) \geq 1-\delta$ and the half-gap condition (24) holds. Then, on $E_T(K;\rho_n)$ (12),*

$$\|\sin\Theta(\mathcal{V}_{K,d}, \widehat{\mathcal{V}}_{K,d})\|_{\mathrm{op}} \leq \frac{2\rho_n(\delta)}{\sqrt{\kappa_0^2 + \gamma(K-K^\star)}}. \tag{25}$$

Proof. Substituting the lower bound in (24) into (22) gives (25). □

A.4 Maximum-Entropy Laws

We recall two standard entropy facts used in the maximum-entropy regularization argument. Let X be an absolutely continuous random vector in $\mathbb{R}^d$ with density p_X. Its differential entropy is

$$h(X) := -\int_{\mathbb{R}^d} p_X(x)\log p_X(x)\,dx,$$

whenever the integral is well-defined.

Lemma A.9 (Gaussian maximum entropy under fixed covariance)**.** *Let X be an absolutely continuous random vector in $\mathbb{R}^d$ with mean μ and covariance $\Sigma \succ 0$. Then*

$$h(X) \le \frac{1}{2}\log\big((2\pi e)^d \det \Sigma\big),$$

with equality if and only if $X \sim \mathcal{N}(\mu, \Sigma)$.

Proof. Let $G \sim \mathcal{N}(\mu, \Sigma)$. Since $D_{\mathrm{KL}}(\mathcal{L}(X) \,\|\, \mathcal{L}(G)) \ge 0$,

$$-h(X) - \mathbb{E}\log p_G(X) \ge 0.$$

Because X and G have the same mean and covariance,

$$-\mathbb{E}\log p_G(X) = \frac{1}{2}\log\big((2\pi)^d \det \Sigma\big) + \frac{1}{2}\mathbb{E}[(X-\mu)^\top \Sigma^{-1}(X-\mu)] = \frac{1}{2}\log\big((2\pi e)^d \det \Sigma\big).$$

Hence the claimed bound follows. Equality holds iff the KL divergence is zero, i.e. iff X has the Gaussian law. □

Lemma A.10 (Trace-constrained maximum entropy)**.** *Let X be an absolutely continuous random vector in $\mathbb{R}^d$ with mean zero, covariance $C \succ 0$, and $\operatorname{tr}(C) = \tau$. Let $h(X)$ denote its differential entropy. Then*

$$h(X) \le \frac{1}{2}\log\big((2\pi e)^d \det C\big) \le \frac{d}{2}\log\left(2\pi e\,\frac{\tau}{d}\right).$$

Equality holds if and only if $X \sim \mathcal{N}(0, \frac{\tau}{d} I_d)$.

Proof. By Lemma A.9, applied with $\Sigma = C$ and $\mu = 0$,

$$h(X) \le \frac{1}{2}\log\big((2\pi e)^d \det C\big),$$

with equality iff $X \sim \mathcal{N}(0, C)$. For the second inequality, arithmetic-geometric mean (AM-GM) inequality gives

$$\det C = \prod_{i=1}^{d} \lambda_i(C) \le \left(\frac{1}{d}\sum_{i=1}^{d}\lambda_i(C)\right)^d = \left(\frac{\tau}{d}\right)^d,$$

with equality iff $\lambda_1(C) = \cdots = \lambda_d(C) = \tau/d$, i.e., $C = \frac{\tau}{d} I_d$. Combining the equality conditions gives $X \sim \mathcal{N}(0, \frac{\tau}{d} I_d)$. □

Connection to predictive isotropy. Theorem 4.2 is the recovery-bound instance of the same fixed-trace principle: at fixed signal energy $\operatorname{tr}(H_K) = \tau$, the flat spectrum $(\tau/d, \ldots, \tau/d)$ is the unique majorization-minimum among positive rank-d spectra with total mass τ [23, 24]. Equivalently, by AM-GM, it maximizes the log-determinant term $\frac{1}{2}\sum_{i=1}^{d}\log\lambda_i(H_K)$ among fixed-trace spectra. The finite-sample condition $\kappa_K^2 \to 1$ is therefore the empirical manifestation of uniform predictive-energy allocation across the rank-d range.

B Related Works and Discussion

Finite-sample theory of joint-embedding learning. Theory for self-supervised representation learning has largely focused on downstream transfer, contrastive objectives, or kernel-regime joint-embedding models. Contrastive analyses give downstream guarantees under latent-class or multi-view assumptions, including labeled-sample-complexity improvements in latent-class settings [25, 26]. Spectral analyses further show how augmentation structure can induce useful representation subspaces with linear-probe guarantees [27]. Related pretext-task and non-contrastive analyses explain how paired views, conditional independence, predictors, and training dynamics support downstream representations and collapse avoidance [28, 29]. Kernel-regime analyses characterize how augmentations, inductive bias, and regularization shape SSL generalization and optimal joint-embedding representations [30, 31]. These results establish finite-sample theory for broad SSL settings, but they do not analyze the JEPA predictive objective as an empirical-risk problem with an explicit low-rank predictor bottleneck. Closer to JEPA, recent deep-linear work explains why latent prediction can avoid noisy reconstruction targets by studying training-dynamics implicit bias [32].

In contrast, our analysis targets the ERM of JEPA itself. Under linearization, the predictive objective becomes reduced-rank regression (Proposition 3.1), yielding a finite-sample risk decomposition into a population floor and a finite-sample spectral excess term (Theorem 3.15).

Encoder isotropy and anti-collapse. A central design principle in non-contrastive SSL is to prevent collapse by controlling the encoder embedding distribution. Whitening-based SSL scatters latent features through batch whitening, Barlow Twins drives the cross-correlation matrix toward identity, and VICReg combines invariance with variance and covariance regularization [16, 33, 6]. Recent JEPA-specific work strengthens this encoder-side perspective by promoting isotropic-Gaussian embeddings as the optimal representation distribution [5, 13].

Our result changes the primitive object. For JEPA, risk is uniquely controlled by the spectrum of the whitened end-to-end map (Lemma 4.1 and Theorem 4.2). Encoder isotropy is therefore a gauge choice in the encoder-predictor factorization satisfying predictive isotropy (Lemma 4.4 and Theorems 4.5 and 4.6).

Entropy and Gaussianity. Our maximum-entropy argument separates the risk condition from its distributional completion. The finite-sample bound requires spectral balancing of the predictive covariance (Theorem 4.2), while Gaussianity enters as the least-committing law satisfying this covariance-level condition. Thus isotropic-Gaussian embeddings arise as the entropy-optimal completion of predictive isotropy, realizable through a suitable encoder-predictor gauge (Lemma 4.4 and Theorem 4.6). This reframes Gaussian embedding regularization as an encoder-side instance of an ERM condition, specifically via our regularizer (18). Stronger distributional constraints should be justified by the loss and task rather than by collapse prevention alone.

Limitations. Our analysis is deliberately scoped to a linearized JEPA model with a frozen target encoder. The linear assumption makes the predictor bottleneck explicit and allows the JEPA objective to reduce to reduced-rank regression, but it does not capture representation-dependent feature learning in deep encoders. Similarly, the frozen-teacher assumption isolates the ERM geometry of the context encoder and predictor, but omits the coupled training dynamics induced by EMA or jointly updated target networks. Therefore, our results should be read as characterizing the local or linearized risk of JEPA, rather than a complete theory of nonlinear dynamics. Extending predictive isotropy to adaptive teachers and nonlinear feature learning remains an important direction for future work.

C Proofs of Main Text

C.1 Proof of Rank of Reduced-Rank Solution

Proof of Lemma 3.2. Using the RRR solution in Proposition 3.1,

$$B_r^\star = B_{\mathrm{OLS}}^{\mathrm{raw}} \Sigma_{ZZ}^{1/2} V_{K,r} V_{K,r}^\top \Sigma_{ZZ}^{-1/2}.$$

Since $\Sigma_{ZZ} \succ 0$, the factor $\Sigma_{ZZ}^{-1/2}$ is invertible, so

$$\operatorname{rank}(B_r^\star) = \operatorname{rank}\left(T_K V_{K,r} V_{K,r}^\top\right).$$

Let $T_K = U_K \operatorname{diag}(\sigma_1, \ldots, \sigma_{\operatorname{rank}(T_K)}, 0, \ldots, 0)\, V_K^\top$ be an SVD. The columns of $V_{K,r}$ are the top-r right singular vectors of T_K. Since V_K has orthonormal columns, $V_{K,r} V_{K,r}^\top$ projects onto the top-r right-singular subspace, so

$$T_K V_{K,r} V_{K,r}^\top = U_K \operatorname{diag}(\sigma_1, \ldots, \sigma_{\min(r, \operatorname{rank}(T_K))}, 0, \ldots, 0)\, V_K^\top,$$

which has rank $\min\{r, \mathrm{rank}(T_K)\}$. This yields (7). $\square$

C.2 Proof of Rank Growth

Proof of Proposition 3.4. From (8),

$$H_{K+1} = H_K + M_{K+1}^\top M_{K+1}, \tag{26}$$

so for every $v \in \mathbb{R}^{D_x}$,

$$v^\top H_{K+1} v = v^\top H_K v + \|M_{K+1} v\|^2.$$

Both terms on the right are nonnegative, so $v^\top H_{K+1} v = 0$ holds if and only if both $v^\top H_K v = 0$ and $M_{K+1} v = 0$. Hence

$$\ker(H_{K+1}) = \ker(H_K) \cap \ker(M_{K+1}) \subseteq \ker(H_K).$$

Therefore

$$\dim\ker(H_{K+1}) \leq \dim\ker(H_K).$$

Since $H_K \in \mathbb{R}^{D_x \times D_x}$, rank-nullity gives

$$\mathrm{rank}(H_{K+1}) = D_x - \dim\ker(H_{K+1}) \geq D_x - \dim\ker(H_K) = \mathrm{rank}(H_K).$$

Using $\mathrm{rank}(H_K) = \mathrm{rank}(T_K) = r_{\mathrm{pop}}(K)$, we obtain $r_{\mathrm{pop}}(K+1) \geq r_{\mathrm{pop}}(K)$, proving (1).

For (2), suppose $\ker(H_K) \not\subseteq \ker(M_{K+1})$ and pick $v \in \ker(H_K) \setminus \ker(M_{K+1})$. Then $v \in \ker(H_K)$ but $v \notin \ker(H_{K+1})$, so the inclusion above is strict, giving $r_{\mathrm{pop}}(K+1) > r_{\mathrm{pop}}(K)$. $\square$

C.3 Proof of Heterogeneity and Identifiability

Proof of Proposition 3.8. On the assumed perturbation event, $\|\widehat{T}_K - T_K\|_{\mathrm{op}} \leq \eta_n^T(\delta)$. Since $\mathrm{rank}(T_K) = r_{\mathrm{pop}}(K)$, applying Lemma A.6 with $r = r_{\mathrm{pop}}(K)$, $\epsilon = \eta_n^T(\delta)$, and Wedin gap $g_{r_{\mathrm{pop}}}(T_K) = \sqrt{\lambda_{\mathrm{het}}}$ (9) yields the claim. $\square$

C.4 Proof of Capacity Lower Bound

Proof of Proposition 3.9. By population least squares, $B_{\mathrm{OLS}}^{\mathrm{raw}} = \Sigma_{YZ}\Sigma_{ZZ}^{-1}$ is the unconstrained minimizer of $\mathbb{E}\|Y - BZ\|_2^2$. Therefore, for every $B \in \mathbb{R}^{Kd \times D_x}$,

$$\mathbb{E}\left[\|Y - BZ\|_2^2\right] = \mathbb{E}\left[\|Y - B_{\mathrm{OLS}}^{\mathrm{raw}} Z\|_2^2\right] + \left\|(B - B_{\mathrm{OLS}}^{\mathrm{raw}})\Sigma_{ZZ}^{1/2}\right\|_F^2.$$

Thus, for any r,

$$\inf_{\mathrm{rank}(B)\leq r} \mathbb{E}\left[\|Y - BZ\|_2^2\right] = \mathbb{E}\left[\|Y - B_{\mathrm{OLS}}^{\mathrm{raw}} Z\|_2^2\right] + \inf_{\mathrm{rank}(B)\leq r} \left\|B\Sigma_{ZZ}^{1/2} - T_K\right\|_F^2.$$

Since $\Sigma_{ZZ}^{1/2}$ is invertible, the rank constraint remains $\mathrm{rank}(B\Sigma_{ZZ}^{1/2}) \leq r$. By the Eckart-Young theorem [34] for Frobenius-norm low-rank approximation,

$$\inf_{\mathrm{rank}(B)\leq r} \left\|B\Sigma_{ZZ}^{1/2} - T_K\right\|_F^2 = \sum_{j>r} \sigma_j(T_K)^2. \tag{27}$$

This proves (10). $\square$

C.5 Proof of Post-Saturation Risk Decomposition

Proof of Proposition 3.11. By definition, $\mathcal{R}_K(\widehat{\mathcal{V}}_{K,d}) = \mathcal{R}^\star_K(d) + \big(\mathcal{R}_K(\widehat{\mathcal{V}}_{K,d}) - \mathcal{R}^\star_K(d)\big)$. The first term equals the underbraced architectural floor by Proposition 3.9. For the second, combining Corollary A.8 with the subspace-Lipschitz Lemma 3.10 yields, with probability at least $1-\delta$,

$$\mathcal{R}_K(\widehat{\mathcal{V}}_{K,d}) - \mathcal{R}^\star_K(d) \le L\,\|\sin\Theta(\mathcal{V}_{K,d}, \widehat{\mathcal{V}}_{K,d})\|_{\mathrm{op}} \le \frac{2L\rho_n(\delta)}{\sqrt{\kappa_0^2 + \gamma(K-K^\star)}}. \tag{28}$$

Adding the two bounds yields (14). ◻

C.6 Proof of Unified Risk Decomposition

Proof of Theorem 3.15. By the reduced-rank regression solution in Lemma A.2, using the whitening argument as in Proposition 3.9, $\mathcal{R}^\star_K(r)$ is exactly the optimal population risk among rank-r end-to-end maps. Next, on $E_T(K;\rho_n)$ (12), the Wedin perturbation bound gives

$$\|\sin\Theta(\mathcal{V}_{K,r}, \widehat{\mathcal{V}}_{K,r})\|_{\mathrm{op}} \le \frac{\rho_n(\delta)\|T_K\|_{\mathrm{op}}}{\Delta^{\mathrm{abs}}_{K,r} - \rho_n(\delta)\|T_K\|_{\mathrm{op}}} = \frac{\rho_n(\delta)}{\bar{\Delta}_{K,r} - \rho_n(\delta)},$$

because $\Delta^{\mathrm{abs}}_{K,r} = \sigma_r(T_K) - \sigma_{r+1}(T_K)$, $\bar{\Delta}_{K,r} = \Delta^{\mathrm{abs}}_{K,r}/\|T_K\|_{\mathrm{op}}$, and $\rho_n(\delta) < \bar{\Delta}_{K,r}$. This is the same perturbation step used in Proposition A.7, now applied at rank r. By Lemma 3.10,

$$\mathcal{R}_K(\widehat{\mathcal{V}}_{K,r}) - \mathcal{R}^\star_K(r) \le L\|\sin\Theta(\mathcal{V}_{K,r}, \widehat{\mathcal{V}}_{K,r})\|_{\mathrm{op}}.$$

Substituting the Wedin bound into this inequality gives (15). ◻

C.7 Proof of Subspace-Lipschitz Excess Risk

Proof of Lemma 3.10. By the whitening reduction in the proof of Proposition 3.9, the per-target risk of a predictive subspace $\mathcal{V}$ decomposes as

$$K\,\mathcal{R}_K(\mathcal{V}) = \mathbb{E}\big[\|Y - B^{\mathrm{raw}}_{\mathrm{OLS}}Z\|^2\big] + \|T_K(I - P_{\mathcal{V}})\|_F^2,$$

where $P_{\mathcal{V}}$ is the orthogonal projection onto $\mathcal{V}$: the first term is the unconstrained OLS floor and the second the residual from restricting the whitened predictor to $\mathcal{V}$. Since $\mathcal{V}_{K,r}$ is the top-r right singular subspace of T_K, Eckart–Young gives $\mathcal{R}^\star_K(r) = \mathcal{R}_K(\mathcal{V}_{K,r}) = \frac{1}{K}\big(\mathbb{E}[\|Y - B^{\mathrm{raw}}_{\mathrm{OLS}}Z\|^2] + \sum_{j>r}\sigma_j(T_K)^2\big)$, the population rank-$r$ floor.

Write $P^\star := P_{\mathcal{V}_{K,r}}$ and $\widehat{P} := P_{\widehat{\mathcal{V}}_{K,r}}$. For any orthogonal projection P, $\|T_K(I-P)\|_F^2 = \mathrm{tr}((I-P)H_K(I-P)) = \mathrm{tr}(H_K) - \mathrm{tr}(H_K P)$, so

$$K\big(\mathcal{R}_K(\widehat{\mathcal{V}}_{K,r}) - \mathcal{R}^\star_K(r)\big) = \mathrm{tr}\big(H_K(P^\star - \widehat{P})\big).$$

Let $H_K = \sum_i \lambda_i(H_K)\,u_i u_i^\top$ with $\lambda_1(H_K) \ge \lambda_2(H_K) \ge \cdots \ge 0$, so $P^\star$ projects onto $u_1, \dots, u_r$. Using $u_i^\top P^\star u_i = 1$ for $i \le r$ (and 0 otherwise) and $u_i^\top \widehat{P} u_i = \|\widehat{P}u_i\|^2$,

$$\begin{aligned}\mathrm{tr}\big(H_K(P^\star - \widehat{P})\big) &= \sum_{i\le r}\lambda_i(H_K)\,\|(I-\widehat{P})u_i\|^2 - \sum_{i>r}\lambda_i(H_K)\,\|\widehat{P}u_i\|^2 \\ &\le \lambda_1(H_K)\sum_{i\le r}\|(I-\widehat{P})u_i\|^2.\end{aligned}$$

The remaining sum equals $\|(I-\widehat{P})P^\star\|_F^2 = \|\sin\Theta(\mathcal{V}_{K,r},\widehat{\mathcal{V}}_{K,r})\|_F^2$, and $\lambda_1(H_K)=\sigma_1(T_K)^2$, so

$$\mathcal{R}_K(\widehat{\mathcal{V}}_{K,r}) - \mathcal{R}_K^\star(r) \le \frac{\sigma_1(T_K)^2}{K}\,\|\sin\Theta(\mathcal{V}_{K,r},\widehat{\mathcal{V}}_{K,r})\|_F^2.$$

Since both subspaces have dimension r, $\|\sin\Theta\|_F^2 \le r\|\sin\Theta\|_{\mathrm{op}}^2 \le r\|\sin\Theta\|_{\mathrm{op}}$, the last step using $\|\sin\Theta\|_{\mathrm{op}} \le 1$. This gives the claim with $L = r\,\sigma_1(T_K)^2/K$. The Frobenius bound is quadratic in $\sin\Theta$; the operator-norm Lipschitz form is the weaker consequence used in Theorem 3.15. ☐

C.8 Proof of Predictive Isotropy

Proof. Let $T_K = U_K S_K V_K^\top$ be a compact singular value decomposition, where $U_K \in \mathbb{R}^{Kd\times d}$ and $V_K \in \mathbb{R}^{D_x\times d}$ have orthonormal columns, and $S_K = \mathrm{diag}(\sigma_1(T_K),\ldots,\sigma_d(T_K))$. Since $\mathrm{rank}(T_K)=d$, $P_K = T_K\widetilde{Z}$ lies in $\mathrm{col}(T_K)=\mathrm{col}(U_K)$. Thus $U_K^\top P_K$ is the intrinsic coordinate vector of P_K on its rank-d predictive range. Covariance isotropy of P_K on this range is therefore equivalent to covariance isotropy of $U_K^\top P_K$ on $\mathbb{R}^d$. Using $P_K = T_K\widetilde{Z}$ and $\mathrm{Cov}(\widetilde{Z}) = I_{D_x}$,

$$\mathrm{Cov}(U_K^\top P_K) = \mathrm{Cov}(U_K^\top T_K\widetilde{Z}) = \mathrm{Cov}(S_K V_K^\top \widetilde{Z}) = S_K^2.$$

Thus P_K has isotropic covariance on its rank-d range if and only if $S_K^2 = cI_d$ for some $c>0$. The nonzero eigenvalues of $H_K = T_K^\top T_K$ are the diagonal entries of S_K^2, so this is exactly (16). ☐

C.9 Proof of Predictive Isotropy Minimize the Excess-Risk Bound

Proof of Theorem 4.2. Let $\lambda_1 \ge \cdots \ge \lambda_d > 0$ denote the nonzero eigenvalues of H_K, with $\sum_{i=1}^d \lambda_i = \tau$. At fixed $\mathrm{tr}(H_K)=\tau$, $\lambda_d \le \tau/d \le \lambda_1$, with both equalities if and only if the spectrum is flat; hence the isotropic spectrum simultaneously minimizes λ_1 and maximizes $\lambda_d = \sigma_d(T_K)^2$, which uniquely maximizes $\kappa_K^2 = \lambda_d/\lambda_1 \le 1$. At rank d, the relative Wedin denominator is $\bar{\Delta}_{K,d} = \sigma_d(T_K)/\|T_K\|_{\mathrm{op}} = \kappa_K$. For a fixed relative perturbation level $\rho_n(\delta)$, the excess-risk factor $\rho_n(\delta)/(\kappa_K - \rho_n(\delta))$ is strictly decreasing in κ_K whenever $\rho_n(\delta) < \kappa_K$. Since the flat spectrum is the unique fixed-trace spectrum with $\kappa_K = 1$, it uniquely minimizes this bound. ☐

C.10 Proof of Covariance Isotropy Subsumption

Proof of Theorem 4.5. By Lemma 4.4, $\Lambda_G(Z) = GV_K^\top \Sigma_{ZZ}^{-1/2} Z$, so

$$\mathrm{Cov}(\Lambda_G(Z)) = GV_K^\top \Sigma_{ZZ}^{-1/2}\,\Sigma_{ZZ}\,\Sigma_{ZZ}^{-1/2} V_K G^\top = GV_K^\top V_K G^\top = GG^\top = I_d,$$

using $V_K^\top V_K = I_d$ from the SVD and $GG^\top = I_d$ by orthogonality property. ☐

C.11 Proof of Linear Gaussian Gauge

Proof of Corollary 4.7. The whitened context $\widetilde{Z} = \Sigma_{ZZ}^{-1/2} Z \sim \mathcal{N}(0, I_{D_x})$ under Gaussian context, and the rows of $V_K^\top$ are orthonormal, so $\Lambda_I Z = V_K^\top \widetilde{Z} \sim \mathcal{N}(0, I_d)$. Hence $G^\star = I$ realizes Theorem 4.6, and the encoder-predictor factorization specializes to (Λ_I, F_I). ☐

D Additional Experimental Details

D.1 Synthetic Experiment Details

Synthetic generator. Each synthetic instance is generated from a latent linear-Gaussian model. We sample a latent vector $s \in \mathbb{R}^{r_\star}$ and observe

$$z = Cs + \sigma_z \epsilon_z, \qquad y_k = a_k^\top s + \sigma_y \epsilon_k,$$

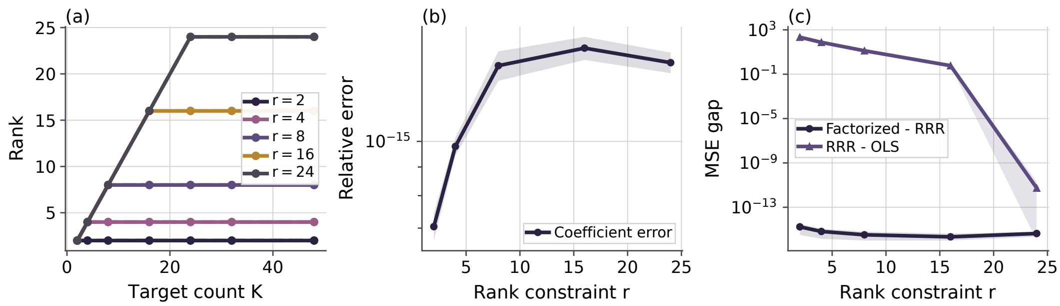

Figure 3: Reduction diagnostics. Panel (a) compares the rank of the closed-form RRR coefficient with $\min\{r, \operatorname{rank}(\Sigma_{YX})\}$; dashed curves are the formula. Panels (b,c) show that an explicit WA factorization reproduces the RRR coefficient and loss up to roundoff.

where $C \in \mathbb{R}^{D_x \times r_\star}$ is a randomly sampled orthonormal context operator, a_k is the kth target direction, and the noise terms are independent isotropic Gaussians. The population predictive structure is known, but the context coordinates are not aligned with the canonical basis.

Target stacks. The target operators are constructed in latent space. In the saturation and bottleneck experiments, we use a complementary target stack whose directions progressively cover the $r_\star$ latent predictive coordinates. In the heterogeneity experiment, the construction parameter is chosen so that the induced population heterogeneity level λ_{het} is sampled approximately evenly. The plot uses the measured heterogeneity level directly on the horizontal axis. In the predictive-isotropy ablation, the target stack is constructed to control the spectrum directly. For a spectrum decay parameter $\rho \in (0, 1]$, we set

$$A^\top A = Q \operatorname{diag}(\lambda_1, \dots, \lambda_d) Q^\top, \qquad \lambda_i = \tau \frac{\rho^{i-1}}{\sum_{j=1}^{d} \rho^{j-1}},$$

where Q is a random orthogonal rotation and τ is fixed across all conditions. This keeps $\operatorname{rank}(A) = d$ and $\operatorname{tr}(A^\top A) = \tau$ fixed, while the measured spectrum condition $\kappa_H^2 = \lambda_d(H_K)/\lambda_1(H_K)$ ranges from 1 to approximately 10^{-8}. Because the context operator has orthonormal columns and σ_z is fixed, the nonzero spectrum of $H_K = T_K^\top T_K$ changes only by the same context-noise scalar across all conditions. The decay parameter ρ is only a generator knob. The figure and interpretation use the measured operator quantities κ_H^2 and entropy gap.

Models and estimators. We first estimate the unrestricted linear predictor by ordinary least squares and then compute its rank-d reduced-rank regression approximation by truncating the singular spectrum. The learned BM-JEPA linear predictor is the best rank-constrained linear map induced by the finite training sample. Population curves use the population cross-covariance operator under the same target stack and bottleneck dimension. Figure 3 gives a direct reduction check for Proposition 3.1 and Lemma 3.2: the population RRR coefficient is explicitly factorized as WA, its rank matches $\min\{r, \operatorname{rank}(\Sigma_{YX})\}$, and the factorized predictor has the same loss as the closed-form RRR solution up to numerical precision.

Metrics. All paper-facing MSE panels report average per-target test mean-squared error, matching the normalization of $\mathcal{R}_K$ in (13). The CSV files also retain fixed-reference-stack diagnostics on a common reference target stack. These are auxiliary representation probes, not the theory-facing risk quantities. In Figure 5, per-target $\mathcal{R}_K$ evaluates the changing target stack used for training. Reference-stack MSE evaluates transfer of the recovered subspace to a common target stack. As heterogeneity increases, $\mathcal{R}_K$ can remain nearly constant or rise slightly because the task becomes less redundant; the reference-stack probe can decrease at the same time because the recovered subspace becomes more complete. The standard-error band of $\mathcal{R}_K$ also shrinks as target coverage spreads across more latent directions, so the per-target average is less dominated by a single shared direction. In Figure 8, panel (d) uses a related reference-stack probe: after estimating the top-d right singular subspace of the whitened operator $\widehat{T}_K$, we fit a linear predictor from that recovered subspace to a fixed flat evaluation target stack. This reference-stack excess MSE is a representation diagnostic. It measures whether the recovered predictive subspace captures all latent directions equally well.

Table 2: Phase-1 synthetic experiment configurations.

Experiment	n_{train}	n_{test}	D_x	$r_\star$	K	d	Noise
K saturation	1024	8192	160	64	sweep	64	0.12
Heterogeneity	32768	8192	64	24	48	24	0.10
Bottleneck	128	8192	64	24	48	sweep	0.16
Unified risk	sweep	8192	80	32	64	16	0.25
Post-saturation	16384	8192	160	64	sweep	64	0.25
Predictive isotropy	1024	8192	64	24	24	24	0.25
Gauge factorization	8192	8192	64	24	24	24	0.25

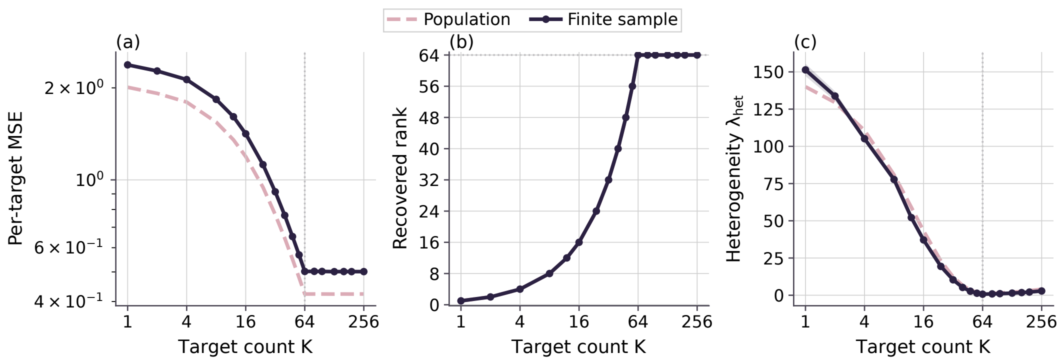

Figure 4: K-saturation diagnostic. Complementary targets reduce per-target MSE and increase recovered rank until predictive rank saturates.

Recovered rank is the effective numerical rank of the learned predictor, computed using a relative singular-value threshold of 0.05 times the leading singular value. The heterogeneity metric λ_{het} is the $r_\star$th nonzero eigenvalue of the target-induced Gram operator. In the latent coordinates this is the same nonzero spectrum as $A^\top A$; in the observed context coordinates it is the same nonzero spectrum as $\Sigma_{ZY}\Sigma_{YZ}$ because $C^\top C = I$. Subspace error is measured by the largest principal-angle sine between the recovered and true predictive subspaces. For the heterogeneity recovery bound, we also compute the empirical perturbation $\widehat{\epsilon} = \|\widehat{\Sigma}_{YZ} - \Sigma_{YZ}\|_{\text{op}}$ and the corresponding Davis-Kahan/Wedin-style denominator $\sqrt{\lambda_{\text{het}}} - \widehat{\epsilon}$. When this denominator is nonpositive, the finite-sample bound is vacuous. We keep these quantities in the CSV as validity diagnostics. We omit them from Figure 5 because the bound is loose for this sweep, and report the direct concentration check separately in Figure 9. The post-saturation experiment plots the recovery-bound diagnostic in a finite window where the relative-conditioning condition is non-vacuous.

For the bottleneck experiment, the plotted spectral-tail quantity is the reduced-rank approximation error $\sum_{j>d}\sigma_j^2$ predicted by the bottleneck spectral-tail result, where $\{\sigma_j\}$ are singular values of the relevant whitened regression operator. In the figure we report the normalized ratio $\sum_{j>d}\sigma_j^2/\sum_j \sigma_j^2$ and compare it with the correspondingly normalized observed excess risk of the rank-d estimator over the unrestricted OLS estimator.

The rank panel in the bottleneck figure reports the effective rank of the learned reduced-rank predictor, not the rank of the stacked cross-covariance. The latter is fixed by the target stack and does not change when the bottleneck dimension is swept. Since the synthetic target stack has latent predictive rank $r_\star = 24$, a well-estimated rank-d predictor is expected to use approximately $\min\{d, r_\star\}$ directions. The rank panel checks that the bottleneck cap is the active rank constraint before $d = r_\star$ and that additional capacity is unused afterward.

Baselines. The main finite-sample baseline is unrestricted OLS, which removes the embedding bottleneck. Population curves are included as reference predictions for the same synthetic distribution. The bottleneck experiment additionally reports a plug-in spectral tail computed from the finite-sample OLS spectrum, so the risk decomposition can be checked using learned empirical quantities rather than only population quantities.

Configuration summary. We list the main Phase-1 configurations used for the reported figures in Table 2.

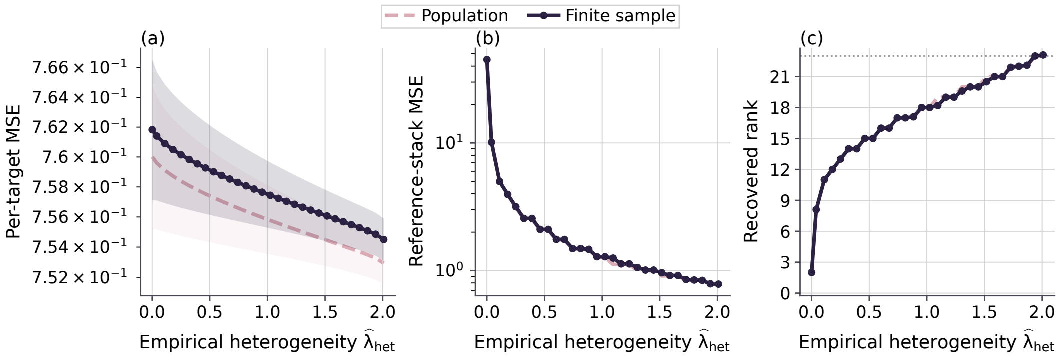

Figure 5: Target-heterogeneity diagnostic. Larger $\widehat{\lambda}_{\text{het}}$ improves rank recovery; the reference-stack probe measures transfer to a common target stack.

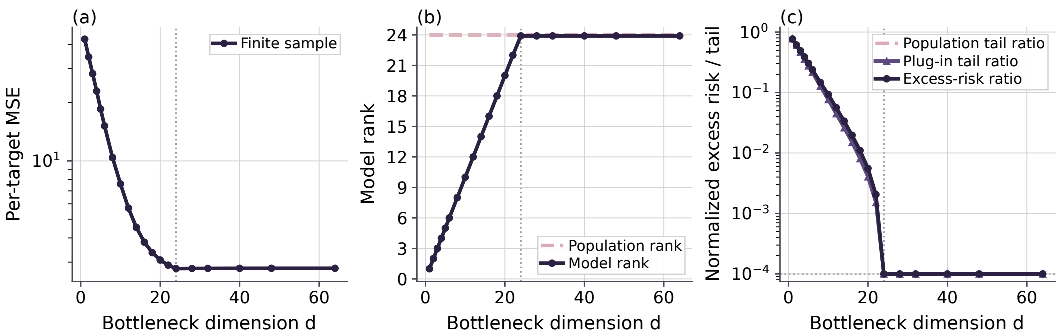

Figure 6: Embedding-bottleneck diagnostic. As d increases, model rank follows $\min\{d, r_{\text{pop}}\}$ and excess risk tracks the spectral-tail ratio.

Additional theory diagnostics. The CSV schema also records the effective rank of the learned RRR coefficient, the finite-sample target value $\min\{d, \widehat{r}_{\text{pop}}(K)\}$ predicted by the rank identity, and their difference. This directly checks the rank identity beyond the displayed rank of the stacked cross-covariance.

Figure 7 gives a diagnostic for Theorem 3.15 separate from the post-saturation special case. The target stack has $K = 64$, $r_\star = 32$, and a controlled rank-r spectral gap at $r = d = 16$, so the spectral-tail term is nonzero while the retained gap is positive. The raw OLS floor is about 0.121, the per-target spectral tail is about 0.244, and the population rank-r risk is therefore about 0.365. We sweep $n_{\text{train}} \in \{512, 1024, 2048, 4096, 8192, 16384\}$ over ten seeds. The retained relative gap is $\Delta_{K,r} = 0.408$. The calibrated $\widehat{\rho}_{0.9}$ decreases from 0.291 at $n = 512$ to 0.049 at $n = 16384$, with $9/10$ perturbation-event coverage at each sample size, so the gap-compatible relative-perturbation condition is non-vacuous throughout this window. The empirical subspace-Lipschitz constant is $\widehat{L} = 7.29 \times 10^{-2}$, computed as the maximum of $[\widehat{\mathcal{R}}_K - \mathcal{R}_K^\star(r)]_+ / \|\sin\Theta_T(\mathcal{V}_{K,r}, \widehat{\mathcal{V}}_{K,r})\|_{\text{op}}$ over all plotted runs. The finite-sample per-target risk decreases from 0.392 to 0.366 and stays below the calibrated theorem bound, while approaching the population rank-r floor.

The diagnostic in Figure 1 tests the post-saturation relative-conditioning mechanism. The target stack is full rank from $K^\star = d = 64$ onward, so additional targets do not change $r_{\text{pop}}(K)$. The relevant post-saturation conditioning ratio is

$$\kappa_K^2 = \frac{\lambda_d(H_K)}{\lambda_1(H_K)}, \qquad H_K = T_K^\top T_K, \quad T_K = \Sigma_{YZ}^{(K)} \Sigma_{ZZ}^{-1/2}.$$

The initial saturated target stack is deliberately imbalanced. Writing $A_K^\top A_K$ for the latent target-stack design Gram, we initialize

$$A_{K^\star}^\top A_{K^\star} = Q \,\mathrm{diag}(c_1, \ldots, c_{64}) Q^\top,$$

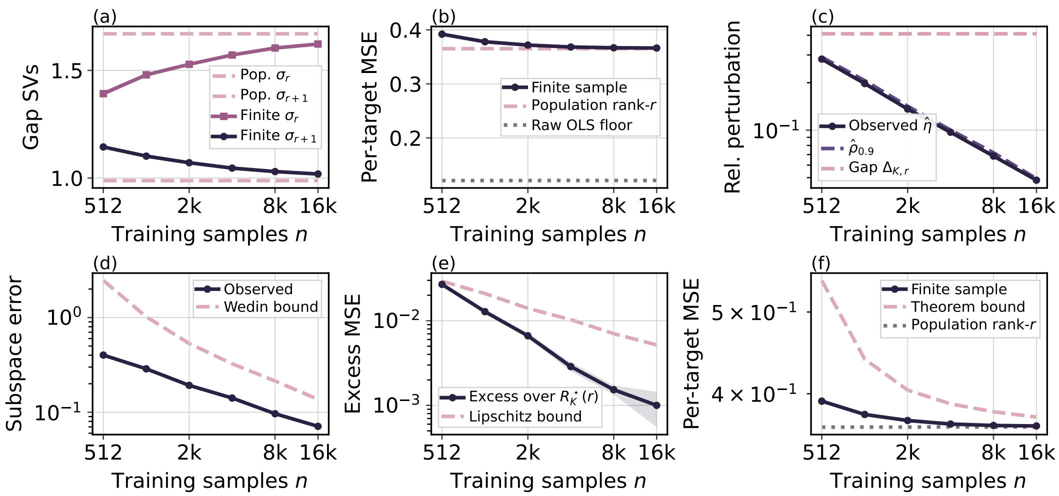

Figure 7: Unified-risk diagnostic for Theorem 3.15. The experiment fixes $r = 16 < r_\star = 32$, so the population spectral tail is active. Panels check the retained gap, perturbation event, Wedin recovery, subspace-Lipschitz bridge, and final risk bound.

where Q is a fixed random orthogonal rotation and

$$c_i = 0.02^{(i-1)/63}.$$

The stack is already rank-64, but $\kappa_{K^\star}^2 \approx 0.02$. For $K > K^\star$, each additional target row is added to the currently weakest eigendirection, with 0.1 units of target energy. This keeps $r_{\mathrm{pop}}(K) = d$ throughout the plotted window while increasing κ_K^2. The finite-sample subspace error and per-target excess MSE decrease over the same window, consistent with the post-saturation claim: extra targets help only when they improve relative conditioning of the already-accessible predictive subspace. Figure 1 reports the weakest and top eigenvalues of $H_K = T_K^\top T_K$, the relative conditioning ratio κ_K^2, the effective rank of T_K, the risk-stability bound, excess risk, and finite-sample risk. The effective-rank panel checks that $r(T_K) := \mathrm{tr}(T_K^\top T_K)/\|T_K\|_{\mathrm{op}}^2$ remains bounded over the finite window required by the post-saturation relative-geometry condition. We compute the empirical risk-stability ratio

$$\widehat{L}_K = \frac{[\widehat{\mathcal{R}}_K - \mathcal{R}_K^\star]_+}{\|\sin\Theta(\mathcal{V}_K, \widehat{\mathcal{V}}_K)\|_{\mathrm{op}}},$$

using per-target excess MSE in the numerator, and set $\widehat{L} = \max_{K,\mathrm{seed}} \widehat{L}_K$. Panel (d) plots the excess-risk left-hand side against $\widehat{L}\|\sin\Theta(\mathcal{V}_K, \widehat{\mathcal{V}}_K)\|_{\mathrm{op}}$, matching the form of the normalized risk-stability condition. Each seed is one deterministic draw of the synthetic data and target stack, and the same $\widehat{L}$ is used across the full plotted window. In the current run this gives $\widehat{L} = 7.07 \times 10^{-3}$.

For the relative-perturbation statement of Proposition 3.11, we set $\delta = 0.1$ and compute

$$\widehat{\eta}_K = \frac{\|\widehat{T}_K - T_K\|_{\mathrm{op}}}{\|T_K\|_{\mathrm{op}}}.$$

The plotted finite-window envelope is the empirical quantile $\widehat{\rho}_{0.9}$ of $\widehat{\eta}_K$ over the plotted (K, seed) pairs. This is a calibrated relative-noise diagnostic. In the current run, $\widehat{\rho}_{0.9} = 8.61 \times 10^{-2}$ and covers $90/100$ relative-perturbation samples by construction. Panel (e) uses the calibrated bound

$$\frac{2\widehat{L}\widehat{\rho}_{0.9}}{\kappa_K}$$

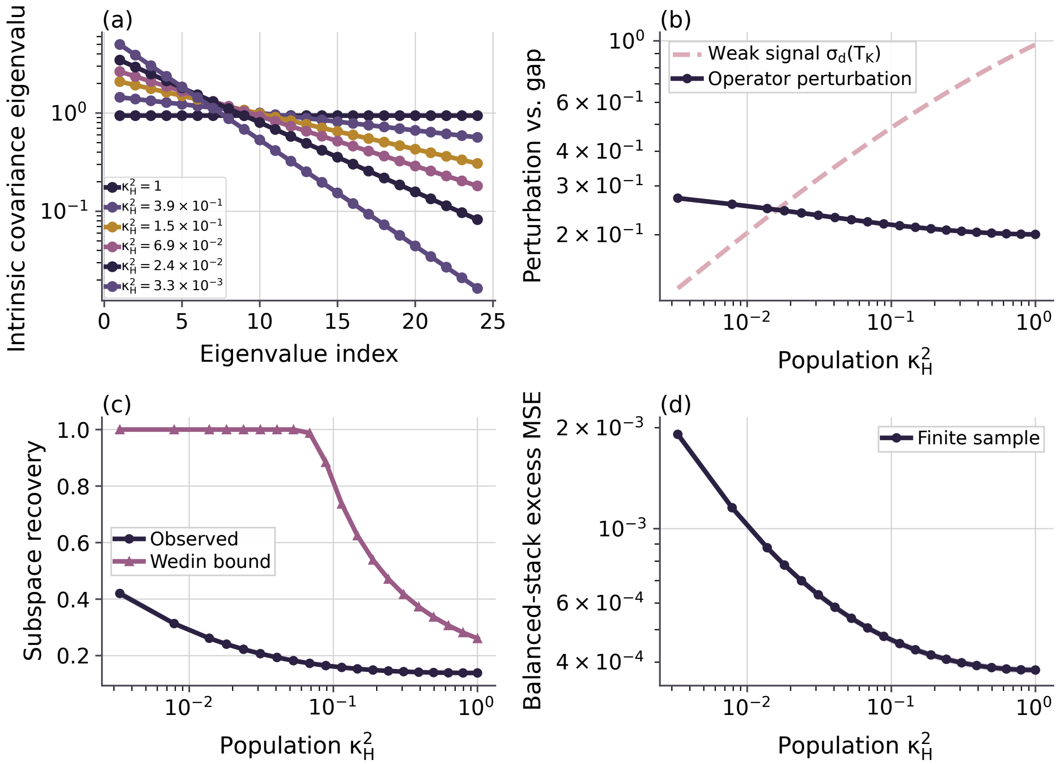

Figure 8: Predictive-isotropy diagnostic. At fixed rank and trace, flatter spectra of $H_K = T_K^\top T_K$ improve recovery and reduce balanced-stack excess MSE.

from the excess-risk term in (14). Panel (f) adds the same term to the population OLS risk estimate, matching the risk-decomposition bound in Proposition 3.11. Here $d = r_{\text{pop}}(K)$, so the spectral-tail term is zero up to numerical precision.

We evaluate the empirical analogue of the half-gap condition stated in Proposition 3.11 using the normalized perturbation

$$\widehat{\eta}_K := \frac{\|\widehat{T}_K - T_K\|_{\text{op}}}{\|T_K\|_{\text{op}}}.$$

The condition $\widehat{\eta}_K \le \kappa_K/2$ holds seed-wise for all deterministic runs in the plotted range. The weakest endpoint, $K = 64$, is deliberately closest to the small-conditioning boundary and remains just inside the half-gap window for the current sample size. Representative values are:

K	$\widehat{\eta}_{K,\text{mean}}$	$\widehat{\eta}_{K,\text{max}}$	$\kappa_K/2$	valid seeds
64	0.0613	0.0656	0.0707	10/10
96	0.0626	0.0688	0.1751	10/10
512	0.0929	0.0947	0.4748	10/10

Concentration and perturbation diagnostics. Figure 9 compares the empirical left-hand sides with calibrated right-hand sides for the concentration and relative-perturbation statements. Panel (a) targets the cross-covariance perturbation controlled by Lemma A.4. Let $U = (Y^\top, Z^\top)^\top$, $M = \mathbb{E}[UU^\top]$, and $r(M) = \text{tr}(M)/\|M\|_{\text{op}}$. The plotted inequality is

$$\|\widehat{\Sigma}_{YZ} - \Sigma_{YZ}\|_{\text{op}} \le \widehat{C}_{0.9}\|M\|_{\text{op}} \left(\sqrt{\frac{r(M) + \log(1/\delta)}{n}} + \frac{r(M) + \log(1/\delta)}{n} \right),$$

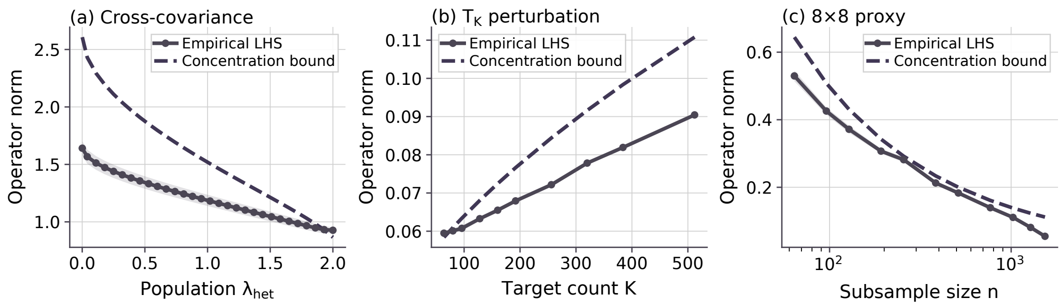

Figure 9: Concentration and perturbation diagnostics. Panels (a)–(c) compare the empirical cross-covariance perturbation, the T_K perturbation, and the real 8×8 digit T_K proxy with their calibrated bounds. Solid curves are empirical left-hand sides; dashed curves are calibrated right-hand sides.

evaluated on the heterogeneity sweep. Panel (b) gives the analogous finite-window perturbation diagnostic with T_K in place of Σ_{YZ}:

$$\|\widehat{T}_K - T_K\|_{\mathrm{op}} \le \widehat{C}_{0.9}\|T_K\|_{\mathrm{op}} \left(\sqrt{\frac{r(T_K) + \log(1/\delta)}{n}} + \frac{r(T_K) + \log(1/\delta)}{n} \right).$$

Panel (c) repeats the T_K calculation on the real 8×8 digit halves. Since the real-data population operator is unknown, we use the full digit dataset to define a reference operator T_{ref} and compare subsample estimates $\widehat{T}_n$ against it. We use $\delta = 0.1$ and set $\widehat{C}_{0.9}$ by a 90% order-statistic calibration.

Predictive-isotropy construction. The fixed-trace predictive-isotropy experiment tests the second-moment prediction in Lemma 4.1 and the fixed-trace bound minimization in Theorem 4.2, with rank and total signal energy held fixed. All conditions use $D_x = 64$, $r_{\mathrm{pop}} = d = K = 24$, $\sigma_z = \sigma_y = 0.25$, and $\mathrm{tr}(H_K) = 22.59$ after the common whitening/context-noise scaling. The generator uses 20 decay values between $\rho = 1.00$ and $\rho = 0.78$, concentrated near the transition where the weakest retained signal becomes comparable to the empirical perturbation scale. These decay values induce a dense sweep of population conditioning values $\kappa_H^2 \in [3.30{\times}10^{-3}, 1]$. Representative points are:

ρ	κ_H^2	entropy gap	$\|\sin\Theta_T\|_{\mathrm{op}}$
1.00	1.0000	0.000	0.139
0.94	$2.41{\times}10^{-1}$	-0.090	0.146
0.90	$8.86{\times}10^{-2}$	-0.253	0.165
0.86	$3.12{\times}10^{-2}$	-0.495	0.207
0.84	$1.81{\times}10^{-2}$	-0.644	0.241
0.78	$3.30{\times}10^{-3}$	-1.191	0.420

The trace column is omitted because it is constant by construction. Since $\widetilde{Z} = \Sigma_{ZZ}^{-1/2} Z$ is Gaussian in this generator, the intrinsic covariance of $P_K = T_K\widetilde{Z}$ on its rank-d range has eigenvalues $\lambda_i(H_K)$. Panel (a) of Figure 8 is the covariance spectrum appearing in Lemma 4.1. Panel (b) plots the two quantities that determine the perturbative denominator: $\sigma_d(T_K)$ and $\|\widehat{T}_K - T_K\|_{\mathrm{op}}$. Panel (c) compares the observed subspace recovery error with the Wedin bound $\|\widehat{T}_K - T_K\|_{\mathrm{op}}/(\sigma_d(T_K) - \|\widehat{T}_K - T_K\|_{\mathrm{op}})$, clipped at one when the denominator is nonpositive. The small-κ_H^2 settings intentionally enter this vacuous-bound regime. These settings mark the transition from perturbative recovery to the boundary where weak directions are hard to recover. In the current ten-seed run, 16 of the 20 spectrum settings have a positive perturbative denominator for every seed, and 10 of 20 satisfy the half-gap diagnostic for every seed.

Gauge-factorization diagnostic. The gauge diagnostic tests Section 4.2. For each seed, we construct the population whitened predictor $T_K = U_K S_K V_K^\top$ and compare identity, random orthogonal, anisotropic diagonal, and nonlinear Gaussianizing gauges under both Gaussian and standardized Laplace SVD-coordinate distributions. For an invertible linear gauge G, the synthetic encoder and

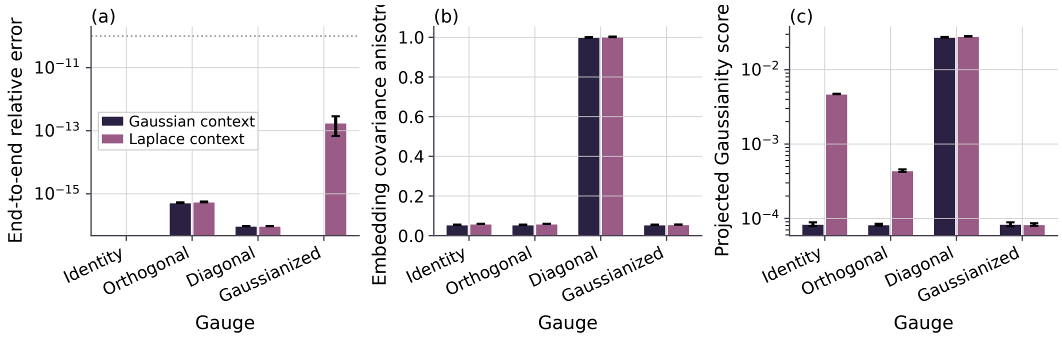

Figure 10: Gauge-factorization diagnostic. Gauge changes preserve prediction while changing encoder covariance and Gaussianity diagnostics.

predictor are

$$\Lambda_G(Z) = GV_K^\top \Sigma_{ZZ}^{-1/2} Z, \qquad F_G(\cdot) = U_K S_K G^{-1}(\cdot).$$

Panel (a) of Figure 10 checks Lemma 4.4: all gauges reproduce the same end-to-end prediction up to numerical precision. Panel (b) checks the covariance part of Theorem 4.5: identity and orthogonal gauges remain covariance-isotropic up to finite-sample error for both context laws, while the diagonal gauge is deliberately anisotropic. Panel (c) reports a Cramer-Wold-style projected standard-normal discrepancy. For Gaussian context, the identity/SVD-aligned gauge already realizes the linear-Gaussian special case in Corollary 4.7. For Laplace context, identity remains covariance-isotropic but not Gaussian; the nonlinear Gaussianizing gauge maps the intrinsic coordinates to standard-normal marginals and sharply reduces the projected discrepancy. This matches the nonlinear gauge construction in Theorem 4.6.

D.2 Real-Image Regularizer Details

Regularization implication. The theory motivates a prediction-aware analogue of embedding Gaussian regularization. Standard embedding regularizers act on the marginal pushforward $\mathcal{L}(AZ)$. The theory-facing object here is the intrinsic predictive pushforward from Lemma 4.1. For a minibatch, project the learned predicted outputs onto the learned rank-d predictive range and collect the intrinsic coordinates into rows $\widehat{p}_n^{\text{int}}$ of

$$\widehat{P}^{\text{int}} \in \mathbb{R}^{N \times d}.$$

The proposed regularizer samples random directions $\mathcal{A} \subset \mathbb{S}^{d-1}$ and averages a univariate Gaussianity test, such as the Epps-Pulley statistic, over the projected samples:

$$\mathcal{R}_{\text{PI}}(\widehat{P}^{\text{int}}; \mathcal{A}) = \frac{1}{|\mathcal{A}|} \sum_{u \in \mathcal{A}} \mathcal{T}\big(\{u^\top \widehat{p}_n^{\text{int}}\}_{n=1}^N\big)\,.$$

This is a Cramer-Wold-style, prediction-aware counterpart to SIGReg: it acts on intrinsic predictor outputs rather than encoder embeddings. Matching projected standard normals is stronger than the second-moment condition of spectral isotropy for the predictive covariance, but it implies the isotropic predictive covariance targeted by Lemma 4.1. The nonlinear diagnostic uses the objective

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{emb}} \mathcal{L}_{\text{emb-iso}} + \lambda_{\text{pred}} \mathcal{R}_{\text{PI}}.$$

The embedding term prevents marginal representation collapse. The prediction term targets the predictive spectrum appearing in the finite-sample recovery and excess-risk bounds. The gauge results explain why a sufficiently strong prediction-aware regularizer can subsume Encoder isotropy at the population level. The combined objective tests whether the Encoder term adds useful finite-sample optimization or collapse-prevention effects.

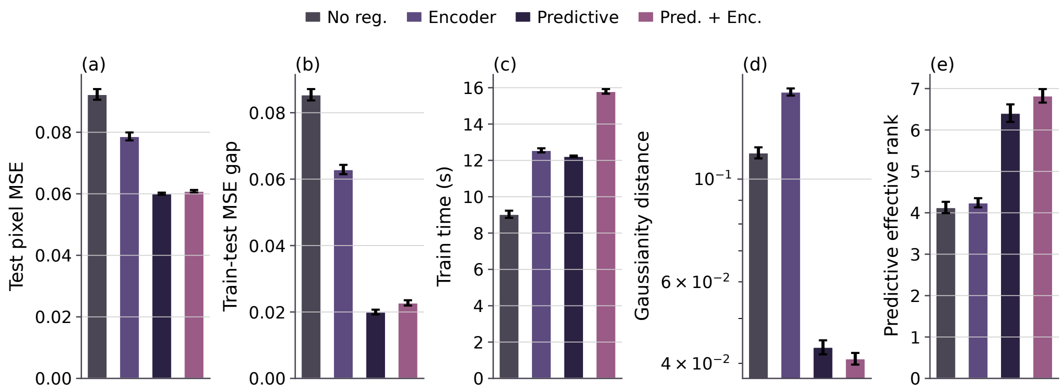

Figure 11: 8×8 digit regularizer diagnostic. Prediction-side regularization improves target-half MSE and prediction-side conditioning.

Real-image regularizer experiments. We run two learned nonlinear left-to-right digit-completion diagnostics. In both cases, the context is the left half of the image and the target is the right half. The small-data `sklearn` task uses 8×8 handwritten digits, an MLP encoder, a 16-dimensional predictor representation, and a linear pixel head. The scaled MNIST task uses 28×28 images, a convolutional encoder, a 128-dimensional predictor representation, and a spatial convolutional decoder. We identify the predictor representation with the finite-network analogue of $\widehat{P}_K^{\mathrm{int}}$ in (18); the pixel head or decoder is the readout used for target prediction.

We compare four objectives: no regularizer, Encoder Gaussianity, prediction-side Gaussianity, and prediction-plus-encoder Gaussianity. The projected-Gaussianity term combines a Cramer-Wold/Epps-Pulley statistic with a covariance-to-identity penalty and is applied either to the encoder embedding or to the predictor representation. The combined condition is a control: if the prediction-side condition captures the risk-relevant pushforward, adding the Encoder auxiliary term should not further improve prediction.

MSE is the mean squared error over right-half target pixels after normalizing image intensities to $[0, 1]$. RMSE is measured on the same gray-scale range. For MNIST, we also report foreground MSE over right-half pixels with intensity larger than 0.05, because background pixels dominate the raw target-half average. The Gaussianity score is a diagnostic distance to projected standard-normal laws, and effective rank is a conditioning diagnostic; the primary outcome is still target prediction risk.

Figures 11 and 12 and Table 1 report the real-image diagnostics. On the low-label 8×8 task, prediction-side regularization reduces test MSE from 9.23×10^{-2} to 6.00×10^{-2} and train-test gap from 8.54×10^{-2} to 2.00×10^{-2}. It reduces predictive Gaussianity distance from 1.14×10^{-1} to 4.33×10^{-2} and raises predictive effective rank from 4.13 to 6.41. The prediction-only and prediction-plus-encoder conditions have nearly identical test MSE; the combined condition is slower.

The MNIST CNN diagnostic uses 10000 training examples, the standard 10000-example MNIST test split, 50 epochs, batch size 256, learning rate 3×10^{-4} with cosine decay, weight decay 10^{-4}, convolutional channels $(64, 128, 256)$, embedding dimension 256, predictor dimension 256, regularizer weight 0.08, covariance weight 1.0, mixed precision when CUDA is available, and 128 random projection directions. In the ten-seed GPU run, prediction-side regularization improves test MSE from 3.68×10^{-2} to 3.43×10^{-2} and foreground MSE from 1.29×10^{-1} to 1.19×10^{-1}. It reduces prediction-side Gaussianity distance from 3.24×10^{-1} to 1.31×10^{-2} and raises predictive effective rank from 5.86 to 10.03. The combined condition gives similar risk and the smallest train-test gap. Figures 2, 13 and 14 show completion examples: the left half of each held-out digit is observed, and each method predicts the right half.

Reproducibility. Synthetic experiments, the 8×8 digit regularizer diagnostic, and the MNIST CNN regularizer diagnostic are run with ten deterministic seeds. The synthetic suite and 8×8 diagnostic can be reproduced with

```
python -m experiments.run_all
```

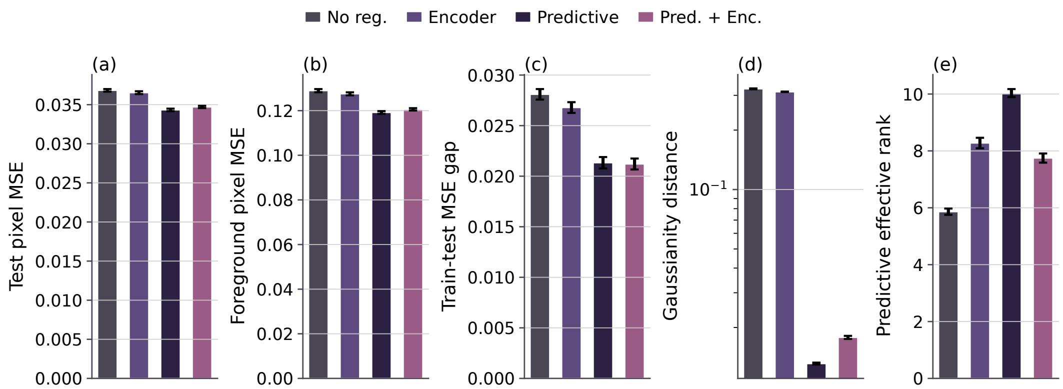

Figure 12: MNIST CNN regularizer diagnostic. The diagnostic reports target-half risk, foreground risk, prediction-side Gaussianity, and training time.

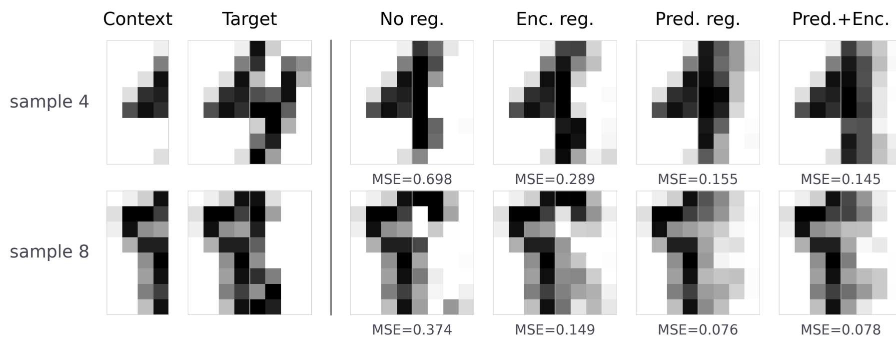

Figure 13: 8×8 digit completion examples. The left half is observed, and each method predicts the right half.

followed by

```
python -m plots.plot_main.
```

The qualitative digit reconstruction figure is generated separately with

```
python -m plots.plot_regularizer_digits_samples.
```

The MNIST diagnostic can also be run directly with

```
python -m experiments.exp_regularizer_mnist_gpu.
```

The scripts write per-experiment CSV files, the combined result table, and the main and appendix PDF figures. Unit tests cover deterministic generation, OLS/RRR behavior, rank constraints, subspace metrics, CSV schema, and figure-reference existence.

Visualization choices. Finite-sample estimates are plotted as solid curves with standard-error bands. Population predictions are plotted as dashed curves with standard-error bands when the population quantity varies across deterministic seeds. Bar charts show standard-error caps and seed-level points. For the bottleneck spectral-tail-ratio panel, values below 10^{-4} are visually floored in the log-scale display to keep the super-threshold regime legible; this affects only the display.

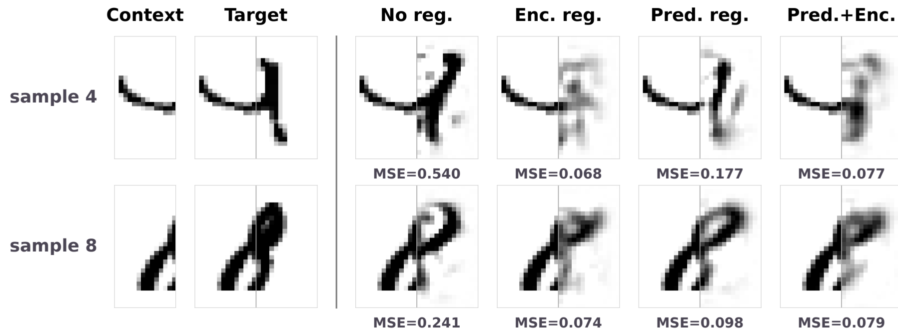

Figure 14: MNIST MLP completion examples. The left half is observed, and each method predicts the right half.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The main claims are scoped to the linearized/frozen-teacher setting and matched to the stated theorems.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Please see the Limitations paragraph at the end of Appendix B (Related Works and Discussion).

   Guidelines: Please see our limitation discussion in appendix.

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.

- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: Please see our main text and appendix.

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Please see our experimental section.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.

(c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: We provide an anonymized supplementary code archive containing experiment scripts, configurations, result CSVs, plotting scripts, tests, and reproduction instructions. The experiments use public datasets (`sklearn` digits and MNIST) and synthetic generators included in the archive.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (https://neurips.cc/public/guides/CodeSubmissionPolicy) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (https://neurips.cc/public/guides/CodeSubmissionPolicy) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Please see our experimental section.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Please see our experimental section.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Please see our experimental section.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics https://neurips.cc/public/EthicsGuidelines?

   Answer: [Yes]

   Justification: All authors read through carefully.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: Please see our impact statement.

    Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper releases no pretrained models, datasets, or other assets that could pose misuse or dual-use risks.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Please see our experimental section.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.

- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The supplementary code archive is documented with a README covering dependencies, reproduction commands, generated outputs, and dataset usage. The paper does not introduce a new dataset or pretrained model.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The paper does not involve crowdsourcing or research with human subjects.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The paper does not involve research with human subjects, so no IRB approval was required.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.

- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: The core contributions of this paper are theoretical (a reduced-rank regression formulation of JEPA, the predictive isotropy condition, and accompanying proofs). LLM-based coding assistants were used only as basic code assistance for implementing the numerical validation experiments (e.g., plotting and data pipelines), and did not impact the core methodology, scientific rigor, or originality of the research. Per NeurIPS policy, declaration is therefore not required.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.